Turning TLC plate photographs into numbers

A complete survey of every route from a phone photo of a TLC plate to (a) a densitogram and spot table, and (b) the text written on the plate — with each route run on our own seven MEHQ plates, and an honest account of where each one breaks.

Prepared for: Evaluation of automated TLC readout
Plates studied: MEHQ-P29, P30, P31, P32, P32-1, P33 (7 unique photographs)
Date: 25 August 2026
Reproducibility: every number in this document comes from code that ran on those seven files. Scripts and raw outputs are attached.

The one-paragraph version. Both halves of this problem are solvable, but not with the photographs we currently take. On the image→graph side the binding constraint is not the algorithm — it is that the phone's auto-exposure has already destroyed the signal on six of the seven plates, and that no solvent front was ever drawn or photographed, which makes R_f mathematically undefined. On the text side, classical and neural OCR fail completely (best character error rate 0.11, on a four-character lane-label row, across 819 runs; not one sample identifier read in full), while a frontier vision-language model reads all seven headers — at a cost of roughly **$1–$4.50 per thousand plates**. The recommendation is a hybrid: fix the capture, use a VLM for everything symbolic (text, lane count, where the annotation bands are), and use deterministic code for everything numeric, reported against a reference lane rather than as a true R_f.

Contents

Part 1 — What we actually need out of a plate photograph

Before comparing methods it is worth being precise about the deliverable, because most of the disagreement between methods turns out to be disagreement about what the answer is supposed to be. A TLC plate is not being photographed for its own sake. It is being photographed to answer a chemical question, and the question determines how much machinery is justified.

1.1 The question the plate is being asked

On these plates the lane labels are consistently `S`, `co`, `R`, `sd` — which on a reaction-monitoring plate normally reads as starting material, co-spot, reaction mixture, and standard. The chemistry question is therefore one of four things, in increasing order of difficulty:

1. **Has the starting material been consumed?** A yes/no about whether the S-lane spot still appears in the R lane. Needs spot presence and lane correspondence. Needs no photometry at all.
2. **Has a new species appeared, and is it the expected one?** Needs spot presence plus a position match between the R lane and the standard lane. Needs relative position, not absolute R_f.
3. **Roughly how far has the reaction gone?** Needs a ratio of intensities between two spots on the same plate. Needs photometry, but only relative photometry.
4. **How much of each species is present?** Needs calibrated photometry against a standard series on the same plate. This is quantitative HPTLC and it is a different experiment from the one these photographs record.

Questions 1 and 2 are what a chemist actually uses a monitoring plate for, and they are comfortably achievable from photographs like ours. Question 3 is achievable with better capture. Question 4 is not achievable from a plate that has no calibration ladder on it, by any method, however sophisticated — that is a chemistry limitation, not a software one, and it is worth saying so plainly before anyone budgets for a model.

1.2 The output contract

Concretely, a useful system should emit, per plate, the following object. Everything in it is either measured, chosen, or flagged — and the design principle that matters most is that the system must never let a *chosen* value masquerade as a *measured* one.

Field	Meaning	Where it can come from	Achievable on our current photos?
sample_id	"MEHQ-P33"	handwriting on the plate, or a LIMS scan	**VLM only** — no OCR engine managed it
conditions	"4 hr (30 + GA...)"	handwriting	**partial** — even a frontier VLM is unsure of the solvent string
n_lanes and **lane_labels**	4; S / co / R / sd	label row, or operator, or signal geometry	**supplied, not measured** — must come from the labels, not the signal
origin_row	where the samples were applied	spotting dots; drawn pencil line	**6 of 7** — ≥2 dots on six plates; plate2 gave one, which our own rule treats as a refusal
front_row	where the solvent stopped	drawn pencil line only	**no** — absent on 7/7 plates
densitogram	optical density vs. distance, per lane	photometry	**1 of 7** — the rest are exposure-clipped
spots[]	position, width, area, uncertainty, confirmed/candidate	peak fitting on the densitogram	**positions yes, areas no**
rf or **rst**	true R_f, or position relative to the standard lane	origin + front, or origin + reference spot	**R_f no** · **R_{st} yes**
flags[]	clipped, plate cut off, no front, mask uncertain…	the pipeline's own diagnostics	**yes** — and this is the most valuable field

1.3 What "convert the image to a graph" means, exactly

There are two different things people mean by this, and conflating them causes most of the confusion in method comparisons.

The 2-D field. Every pixel is converted from a camera value into a physically meaningful quantity — optical density, $OD = \log_{10}(I_0/I)$, where I_0 is what the plate would have read with nothing on it. This grid *is* the graph in the sense that matters: it is exposure-invariant, it is the thing that can be compared between two photographs of the same plate, and every downstream number is derived from it. Getting this right is 80% of the engineering.

The 1-D densitogram. Each lane is collapsed to a curve of OD against migration distance — the familiar chromatogram trace. This is a *presentation* of the 2-D field, and it involves choices (lane width, whether to average or sum, what to do about tilt) that are not forced by the data. Two competent engineers will produce visibly different densitograms from the same plate and both will be right.

1.4 The accuracy that would make this useful

Worth stating up front, because "is it accurate enough?" has no answer without a target. Published reference points: manual R_f reading between analysts agrees to roughly ±0.02–0.05; a fully classical, ICH-validated smartphone TLC pipeline (TLCyzer, *Scientific Reports* 12, 2022) achieves repeatability of **2.79% RSD** and intermediate precision of **4.46% RSD** on quantitation, with recovery 96.8–103.9%. That is the bar a classical method has already cleared — on properly captured images with a calibration series.

> **Targets we would propose for Mstack.** For reaction monitoring: spot detection sensitivity ≥95% for spots a chemist can see, false-positive rate <0.2 spots per plate, position reproducibility ±0.02 in R_{st} units, and — most important — an explicit "insufficient image quality" verdict rather than a silent guess. Quantitation targets should not be set until a calibration ladder is being spotted on the plates, because until then there is nothing to be accurate *to*.

Part 2 — The problem inventory

"Convert the image to a graph" hides at least twelve distinct sub-problems, and they do not have equal weight. Some are solved textbook geometry. Two of them are, on our current photographs, unsolvable in principle. This section walks each one and shows it on our own plates, because an abstract list of difficulties is easy to wave away and a measured one is not.

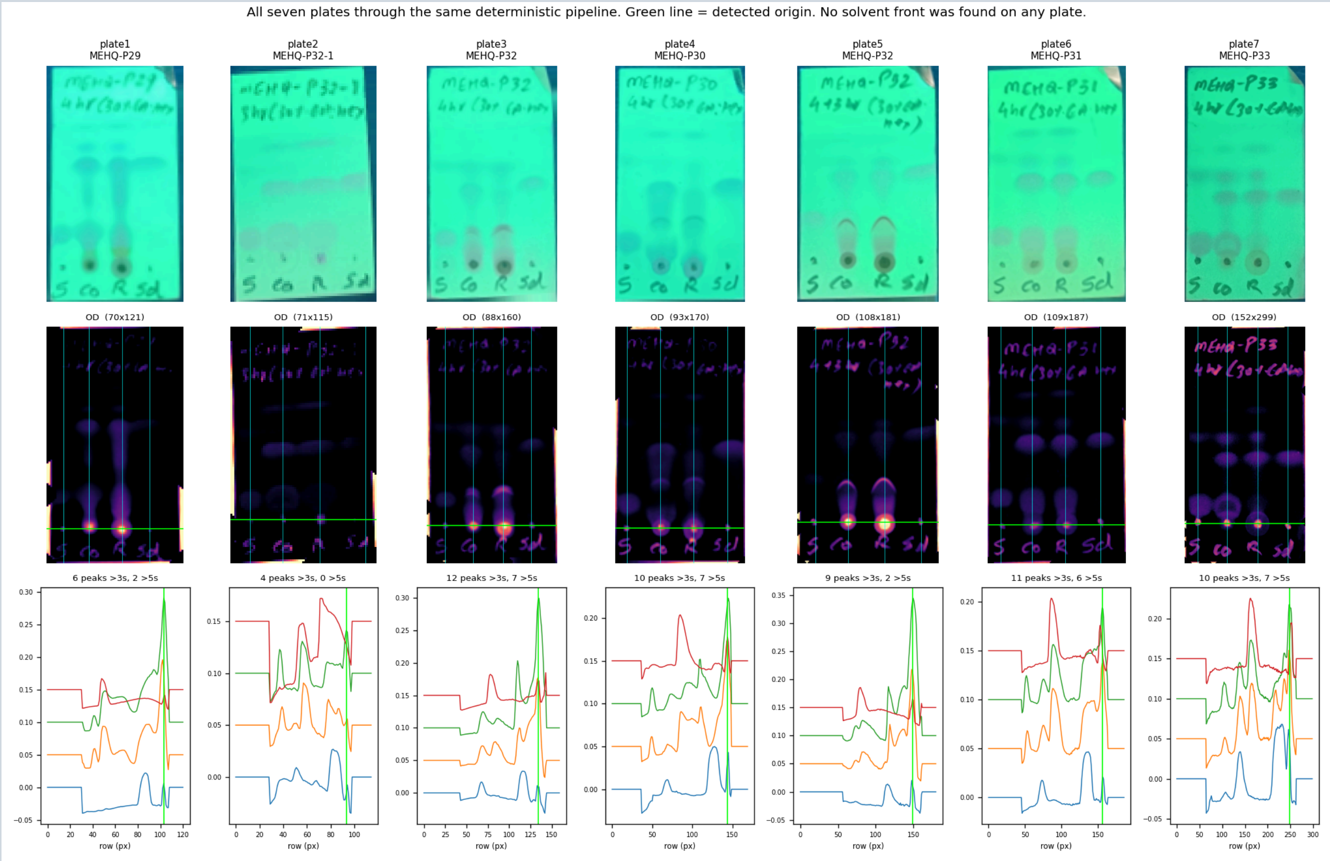

Figure 1. All seven plates through one deterministic pipeline. Top: rectified photograph. Middle: optical-density map with detected lanes (cyan) and the detected origin row (green). Bottom: the four per-lane densitograms. Note in the middle row that the *handwriting is as strong a feature as the chemistry* — the pink text in the OD maps is exactly what a spot detector is designed to find.

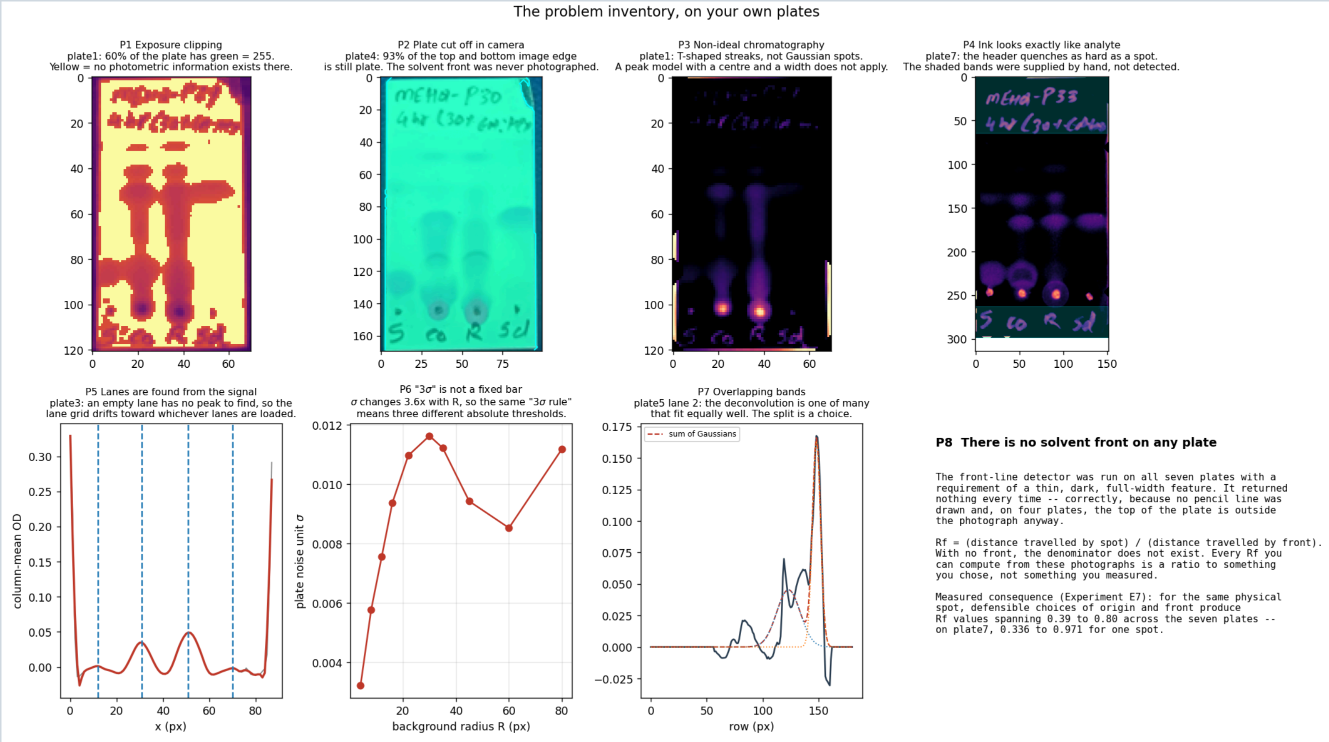

Figure 2. Eight of the twelve problems, each shown on the plate that displays it most clearly.

P1. Exposure clipping — the one that actually decides this project

Under 254 nm the F254 indicator emits green. The green channel is therefore the only channel carrying the quenching signal; red and blue carry noise. On six of our seven plates the phone's auto-exposure has driven the green channel to its maximum value across a large fraction of the plate:

Plate	Photo size (px)	px per lane rectified width / 4	Green channel at 255	Tonal range (p99–p01)	Tilt	Plate runs off frame?
plate1 · MEHQ-P29	71 × 130	17.5	**59.6%**	135	2.2°	no
plate2 · MEHQ-P32-1	77 × 125	17.8	14.4%	157	3.6°	no
plate3 · MEHQ-P32	93 × 166	22.0	**35.2%**	116	4.1°	yes (85% of edge)
plate4 · MEHQ-P30	100 × 170	23.2	22.7%	118	0.8°	yes (93%)
plate5 · MEHQ-P32	112 × 184	27.0	24.0%	137	2.0°	yes (93%)
plate6 · MEHQ-P31	121 × 197	27.2	20.3%	78	3.0°	no
plate7 · MEHQ-P33	158 × 299	38.0	**0.0%**	110	0.4°	yes (77%)

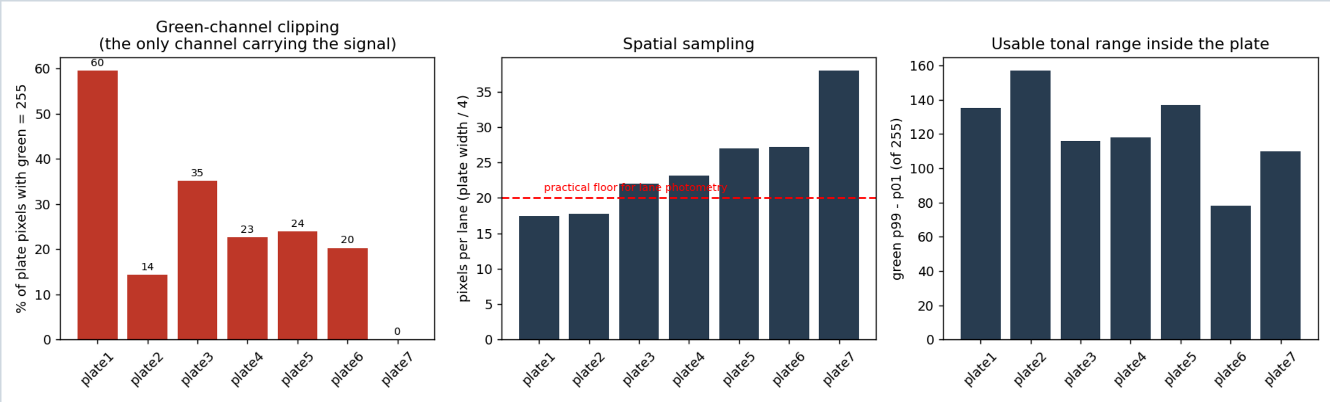

Figure 3. The three capture variables that bound what any method can do. The red line in the centre panel is a practical floor of 20 pixels per lane for lane photometry: plates 1 and 2 fall below it and plate 3 is marginal. Only plate 7 is comfortably clear on all three counts.

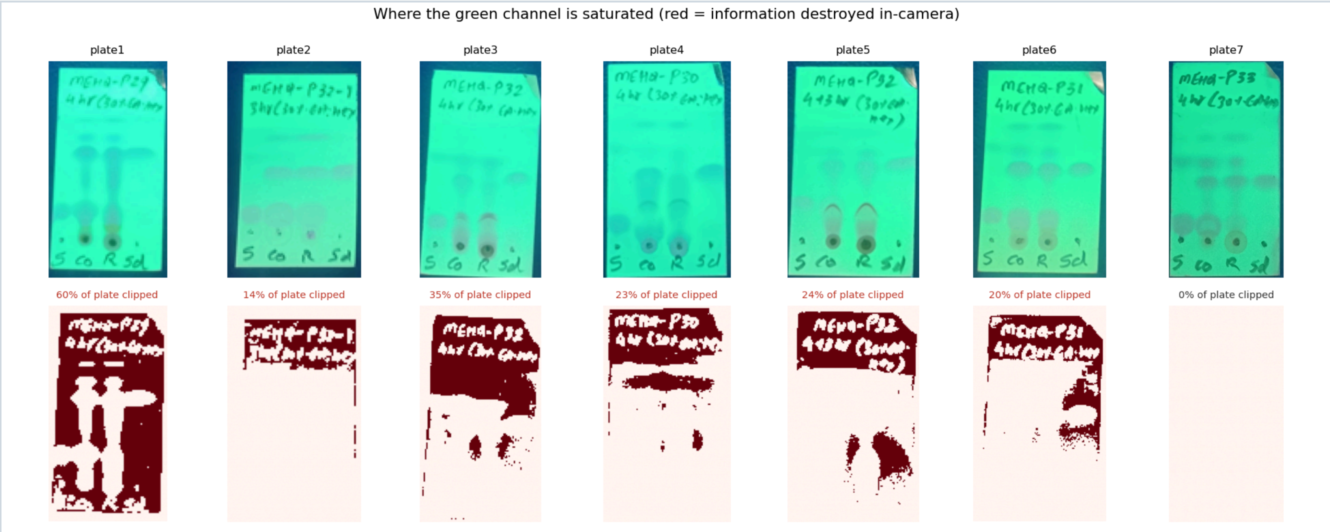

Figure 3b. Red marks every pixel where the green channel reads 255. In those pixels the true brightness could have been 255 or 400 or 4000 — the sensor saturated and the information was discarded before the file was written. No algorithm recovers it.

Why this outranks every algorithmic question. Optical density is $\log_{10}(I_0/I)$. If I_0 — the blank-plate reference — is clipped, then the denominator of every ratio on that plate is wrong by an unknown amount, and it is wrong *non-uniformly*, because clipping happens where the plate is brightest. A faint spot sitting in a clipped region does not read as faint; it reads as whatever the difference is between 255 and the spot, which bears no fixed relationship to how much material is there. On plate1 that describes 60% of the plate.

Concretely: a spot detector will still find the strong spots, and their *positions* will still be right. But every intensity, area and ratio computed from plates 1–6 is uninterpretable, and the sensitivity to faint spots is silently reduced wherever clipping occurs. This is why "which background algorithm is best" is the wrong first question.

P2. Resolution — a real constraint, but not the one you would guess

The images are 0.009–0.047 megapixels; a lane is 17–38 pixels wide and the handwriting is 3–16 pixels tall (measured in §4.1). The intuitive conclusion is that everything degrades. The measurement says otherwise, and the distinction matters for where to spend effort.

We took plate7 (the only clean plate) and downsampled it progressively, re-running the identical pipeline at each scale:

Scale	Rectified plate (px)	px / lane	Candidates (3σ prominence)	Confirmed (>5σ amplitude)	Origin dots found	Apparent R_f of confirmed spots
1.00	152×299	38.0	10	7	4 / 4	0.010 0.010 0.029 0.116 0.446 0.455 0.456
0.75	113×224	28.2	10	6	4 / 4	0.012 0.012 0.037 0.118 0.448 0.457
0.50	76×150	19.0	9	6	4 / 4	0.013 0.048 0.117 0.450 0.458 0.459
0.35	55×104	13.8	9	6	4 / 4	0.006 0.064 0.115 0.446 0.451 0.452
0.25	40×75	10.0	8	6	2 / 4	0.003 0.071 0.117 0.447 0.448 0.453

Spot *positions* are remarkably robust: the same band sits at R_f 0.446–0.459 across a 15-fold reduction in pixel count. That is because a TLC spot is a large, soft object — its centroid is an average over hundreds of pixels, so it survives downsampling. What degrades is **sensitivity** (peaks drop 10→8) and, critically, the **origin-dot anchor**: below about 12 px per lane the small spotting dots stop being resolvable, and at that point the whole migration scale loses its reference.

> **Conclusion to carry forward.** For the image→graph track, resolution is a secondary constraint and our current images are marginally adequate for spot *positions*. For the text track it is the primary and fatal constraint — see Part 4. Resolution and exposure are different problems with different fixes, and only one of them is urgent for the graph.
>
> **With one important qualification.** That downsampling curve was run on plate7, which is our *best* plate: 38 px per lane and the only one without clipping. The other six sit at 17.5–27.2 px per lane — that is, they already start between plate7's 0.5× and 0.75× rungs, and plates 1 and 2 fall below the 20 px-per-lane line drawn in Figure 3. plate2, at 17.8 px per lane, resolved only one of four origin dots. We cannot separate "its dots are faint" from "its dots are unresolved", and the honest statement is that **resolution is comfortably adequate on plate7 and untested-to-marginal on the rest**.

P3. Is cropping even needed? — finding the plate

Yes, and it is cheap. The plate is the largest bright, strongly-green region; a hue-plus-value mask followed by largest-connected-component finds it on all seven plates, covering 79.5–93.2% of the frame. The naive version of this — hue alone — fails, because the dark teal bench surface under UV falls in the same hue band as the plate; adding an Otsu threshold on brightness separates them cleanly. That is a two-line fix but it is the kind of thing that quietly ruins a pipeline if skipped.

The genuinely important part is not finding the plate but **checking whether the plate is entirely inside the photograph**. We measure what fraction of the three-pixel band along each image border still falls inside the plate mask; if most of it does, the plate continues past the frame. On **four of seven plates** that figure exceeds 0.77 on at least one edge — and which edge it is matters:

Plate	top	bottom	left	right	Consequence
plate3	0.00	0.85	0.20	0.33	origin end cropped
plate4	0.85	0.93	0.00	0.00	**both ends cropped**
plate5	0.00	0.93	0.22	0.19	origin end cropped
plate7	0.77	0.53	0.19	0.00	front end cropped

On plates 4 and 7 the top of the plate — where a solvent front would be — was never photographed. On plates 3 and 5 it is the bottom, below the label row, that runs off. A pipeline that silently crops to the visible region produces a confident, wrong migration scale either way.

P4. Tilt and rectification — solved, but with a sting

Tilt is 0.4–4.1° across our plates. Correcting it is standard: take the four corners of the detected plate and apply a projective warp so a vertical distance means the same thing everywhere. This is textbook and it works.

The sting is not in the geometry. It is in what a one-degree change does to *which spot the pipeline decides to report*. We rotated plate7 synthetically from 0° to 12° and re-ran everything, tracking "the highest confirmed spot in the R lane":

Tilt	Plate height after warp	Confirmed peaks (rows)	Reported apparent R_f	R_f of the same physical band
0°	299	`164.8, 246.4`	0.456	0.456
1°	300	**`139.9`**`, 166.0, 246.7`	0.596	0.453
2°	301	`140.6, 166.7, 247.5`	0.594	0.452

The plate extent changed by one pixel — far too little to move R_f by 0.14. What actually happened is that a **third, marginal peak at row ~140 crossed the 5σ bar** and became the tracked spot. The physical band we meant to follow moved 0.456 → 0.453 → 0.452: *stable to 0.004*. Rectification works exactly as advertised.

That is a more useful lesson than the one we went looking for. The geometry is fine; the **reporting rule** is fragile. Any convention of the form "the highest confirmed spot" or "the strongest peak" will silently flip between physically different features whenever a marginal peak crosses the threshold. Report spots by matched position across a plate series, never by rank.

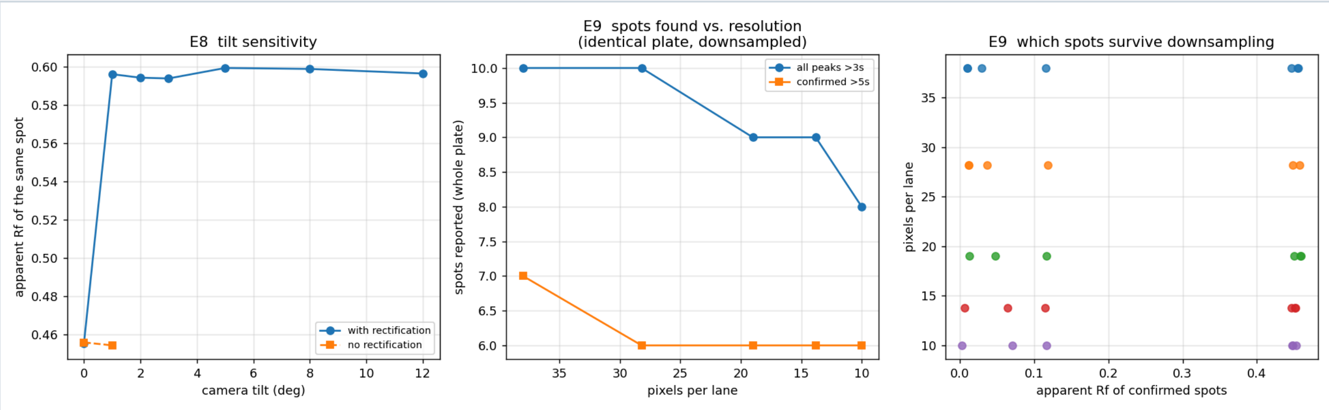

Figure 4. Left: the reported apparent R_f against camera tilt. The step at 1° is a threshold-crossing artefact in the *reporting rule*, not a geometric error — the band itself is stable to 0.004. Centre and right: what downsampling costs.

P5. The illumination problem — and the phantom spots it creates

This is the concern that prompted this study and it deserves the most space. Under a UV cabinet the plate is not evenly lit: on plate7 the fitted illumination surface runs from 0.753 to 0.891 in normalised green — a **15.5% swing** from the bright centre-right to the dark corners (Figure 5, lower left). Without correcting for it, the dark corner of a plate looks like a giant spot. So every serious method estimates I_0, the "what this plate would look like empty" surface, and divides by it.

The problem is that I_0 is not measured. It is inferred from the same pixels that contain the spots, and the inference has a free parameter: how large a region counts as "lighting" rather than "sample". That parameter is a radius, in pixels, chosen by whoever wrote the code.

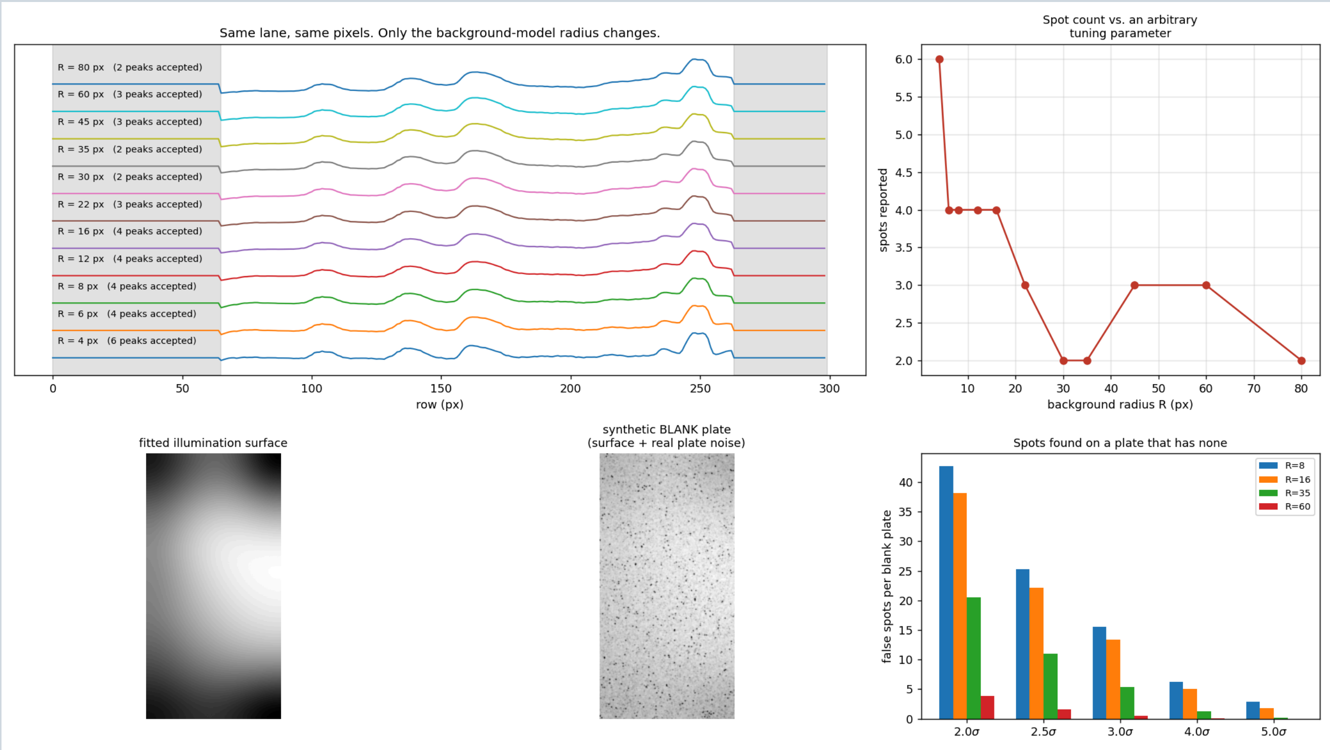

Figure 5. Top left: the same lane of the same plate, processed identically except for the background radius R. Top right: the number of spots the pipeline reports, as a function of that one arbitrary parameter — it ranges from 2 to 6. Bottom right: spots found on a plate that provably has none.

The parameter sweep

Sweeping R from 4 px to 80 px on one lane of plate7, the reported spot count goes `6 → 4 → 4 → 4 → 4 → 3 → 2 → 2 → 3 → 3 → 2`. Nothing about the plate changed. A specific example: a feature at row ~105 is reported as a spot for R ≤ 16 px and absorbed into the background for R ≥ 22 px. It is either a real faint band or model ringing, and the pipeline cannot tell you which.

The noise unit is circular

A design that says "flag anything above 3× this plate's own noise" sounds principled and self-calibrating. It is not, because the noise is measured on the residual *after* background subtraction, so it inherits the background model's arbitrariness. Measured on plate7: σ ranges from **0.0032 at R=4 to 0.0116 at R=30 — a factor of 3.6**. "Three sigma" therefore means three different absolute thresholds depending on a parameter that has nothing to do with the chemistry.

Worse, there is a second circularity. The iterative background model masks out pixels it has decided are spots, and the noise is then measured on what remains. Excluding the most variable pixels lowers the estimate. Measured: σ with spot-masking = **0.01122**; without = **0.01881**. Masking makes every downstream threshold **1.68× more permissive** than it appears to be.

The null test: how many spots does the pipeline find on a plate with no spots?

This is the direct answer to "could we be inventing spots?". We built plates that provably contain none, in two independent ways, and ran the identical detector on them.

- **Null A (synthetic).** Take the fitted illumination surface from plate7 and add noise resampled from a visually blank band of the real plate. 40 realisations.
- **Null B (real texture).** Take the actual residual field from a blank band of plate7 — preserving its true spatial correlation, JPEG texture and all — and tile it mirror-wise over the illumination surface. 13 independent phase shifts. This one cannot be accused of getting the noise statistics wrong, because it *is* the plate's own noise.

Threshold	R=8 px	R=16 px	R=35 px	R=60 px
Null A — mean false spots per blank plate				
2σ	42.7	38.2	20.5	3.9
3σ	15.5	13.4	5.4	0.50
5σ	2.9	1.8	0.23	0.00
Null B — mean false spots per blank plate				
2σ	12.2	11.5	6.0	6.1
3σ	11.5	11.2	5.5	3.0
5σ	11.5	5.7	0.69	0.62
Null B — % of blank plates with ≥1 false spot at 3σ	100%	100%	100%	100%

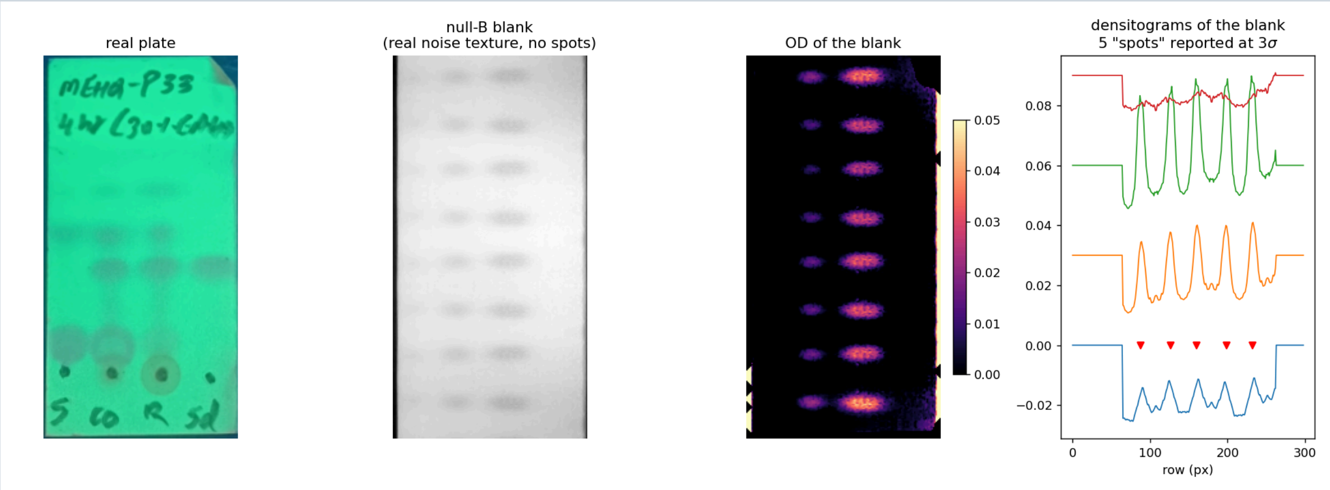

Figure 6. Null test B. Left to right: the real plate; a blank plate built from the real plate's own noise texture; its optical-density map; and its four densitograms, on which the standard 3σ detector reports **five spots** — at rows 88, 127, 160, 199 and 232. There is no chemistry on this plate at all.

> **The answer to the question, with a number on it.** At the conventional operating point (R = 35 px, 3σ) the pipeline reports **5.5 spots on a plate that contains none**, and **100% of blank plates produce at least one**. Even at 5σ — a threshold strict enough to lose genuinely faint bands — 69% of blank plates still yield a false spot at that radius.
>
> This is not a bug in the implementation. It is intrinsic to the approach: you are fitting a smooth surface to noisy data and calling the residual "signal", so wherever the surface happens to sit slightly high, you get a spot. The mitigations are real but partial: a larger radius, a stricter threshold, a minimum spot width, a requirement that a feature appear in more than one lane or in a repeat photograph — and above all, **reporting how many pipelines agree rather than reporting one number** (see Approach 14 in Part 3).

P6. Lanes

Lanes are located by projecting the OD field down each column and finding the strongest vertical bands. This works when every lane is loaded. It fails in a specific and common way: **an empty lane has no signal to find**, so the lane grid slides toward whichever lanes carry material. On plate7 the detected lane centres were x = 16, 53, 92, 130, but the "S" lane is nearly empty and its detected centre sits visibly left of the spotting column in the photograph — enough, at 38 px per lane, to integrate part of the neighbouring lane. (The offset is judged against the image by eye; nothing in the pipeline records where a lane *should* be, which is precisely the gap.)

The robust fix is not a better peak-finder. It is to take the lane count and approximate positions from the *label row* — which is exactly the kind of symbolic reading a VLM does well and a signal-processing pipeline cannot do at all — and then refine within a narrow window. This is the first place where a hybrid design earns its keep.

P7. The origin line

Good news, and the one place these photographs are unambiguously well-suited: the spotting dots are visible on six of the seven plates, and a small-scale blob detector finds them reliably: 4/4 lanes on five plates, 3/4 on plate1, and only 1/4 on plate2 — which under the refusal rule below means plate2 has *no* usable origin, and at 17.8 px per lane we cannot tell whether its dots are genuinely faint or simply unresolved. Taking the median dot row as the origin gives a stable, physically meaningful anchor. It must never be allowed to fall below the label row, and the detector should refuse rather than guess when fewer than two dots are found.

P8. The solvent front — why R_f is undefined here

R_f = (distance travelled by the spot) / (distance travelled by the solvent front). A front-line detector requiring a thin, dark, full-width feature was run on all seven plates. It returned **nothing, on every plate** — correctly, because no pencil front line was drawn, and on four plates the top of the plate is outside the photograph regardless.

The denominator does not exist. Anything called "R_f" computed from these images is a ratio to something that was chosen, not measured. To quantify how much that choice matters, we took one migrated spot per plate and computed R_f under every combination of three defensible origin conventions and four defensible front conventions. Because the pencil-line front is absent everywhere, nine of the twelve combinations yield a value — six on plate5, where three give a negative R_f and are discarded. plate2 is absent from the table because no spot on it cleared the detection threshold at all:

Plate	Spot row	Lowest R_f	Highest R_f	Spread
plate1	99.0	0.039	0.840	0.80
plate3	113.8	0.150	0.539	0.39
plate4	108.0	0.251	0.709	0.46
plate5	149.4	0.078	0.852	0.77
plate6	90.9	0.420	0.937	0.52
plate7	164.8	0.336	0.971	0.64

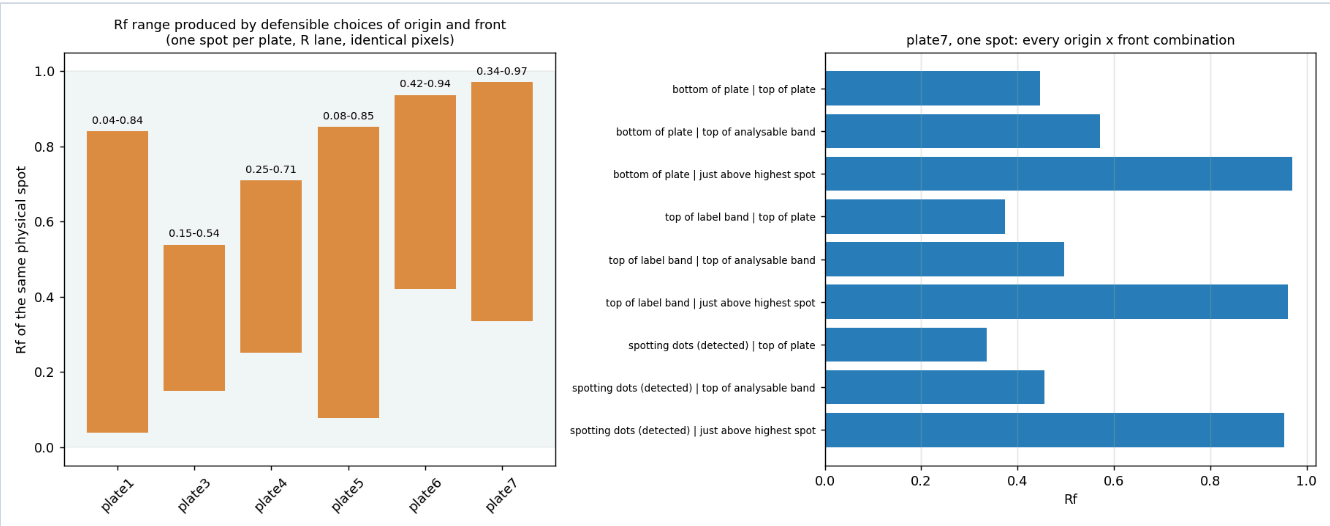

Figure 7. Left: the R_f range each plate's strongest migrated spot can legitimately be assigned. Right: every origin×front combination for one spot on plate7 — the same spot is anywhere from R_f 0.34 to 0.97 depending on conventions no one wrote down.

The fix costs nothing and should be adopted regardless of which method wins. Stop reporting R_f. Report $\mathbf{R_{st}}$ — the spot's migration distance divided by the migration distance of the reference spot in the standard lane *on the same plate*. It needs no front line, it cancels most run-to-run variation in chamber saturation and layer activity, and it is the quantity a chemist actually compares between plates. If a true R_f is required for a report, the only route is to draw and photograph the front line — a pencil mark and two seconds of a chemist's time.

P9. Spot shape: nothing here is a Gaussian dot

Plate1 shows T-shaped streaks running the full height of two lanes. Plates 3, 4 and 5 show comet tailing from overloaded origin spots. Plate7 shows a halo around the R-lane origin dot where the spotting solvent spread. Plate2's spots are so diffuse that the pipeline confirmed **zero** spots at 5σ.

This breaks the standard peak model in two ways. First, a symmetric Gaussian is the wrong function; the chromatographic literature uses the exponentially-modified Gaussian for tailing, and even that will not fit a streak. Second, and more fundamentally, "the position of the spot" stops being well-defined for a streak — is it the leading edge, the centroid, or the mode? Each gives a different R_{st}, and the difference is larger than the measurement precision.

Practical consequence: a streaking lane should be *flagged and not quantified*, not fitted with a model that does not apply. Streak detection is easy — the ratio of the fitted width to the lane width, or the second moment of the lane profile — and it is the kind of honest refusal that makes a system trustworthy.

P10. Overlapping spots and the deconvolution illusion

Where two bands partly merge, a sum-of-Gaussians fit will happily separate them, report two areas, and give both error bars. Those error bars describe how well the model fits, not whether the model is right; a two-peak and a three-peak fit to the same merged band are often statistically indistinguishable. Figure 2, panel 7 shows this on plate5 lane 2. The number of components is a choice being presented as a measurement.

P11. Annotations: ink and analyte are the same thing to a detector

This is the failure that surprised us most. The proposed discriminator — "handwriting is sharp, spots are soft, so a ridge filter separates them" — is scale-dependent, and at our resolution a pen stroke is 1–2 pixels wide, which is exactly the noise scale. Measured: the ridge-based stroke mask fired on **47–71% of all plate pixels**. It is not a weak discriminator here; it carries no information at all.

The darkness-based alternative fails too, for a different reason. Measuring plate7's OD map over all pixels above 3σ: the header band runs p10–p90 = **0.053–0.162** with a maximum of 0.382; the chemistry band runs p10–p90 = **0.038–0.092** with a maximum of 0.352. The distributions overlap almost entirely, and the ink in fact reaches *higher* optical density than the chemistry does. The iterative background model then makes it worse: with a radius large enough to model the lighting (35 px), the model absorbs the handwriting into the "background" over three iterations, which both hides the text and depresses the background under it, biasing everything nearby.

For this report the annotation bands were supplied **by hand**, as fractions of plate height, because no deterministic detector we tried could find them. That is not a gap in effort; it is a structural point. Deciding "this is writing, that is chemistry" is a semantic judgement, and the right tool for a semantic judgement is a model that understands what writing is.

P12. Photometry: OD, Kubelka–Munk, or neither

Silica is a scattering layer, so the physically correct reflectance transform is the Kubelka–Munk function, `KM = (1−R)²/2R`, rather than the absorbance-like `log(I₀/I)`. In practice, on faint quenching zones where R is close to 1, KM is numerically hostile: our M8 variant reported **9 peaks** on a lane where the OD variants reported 2, because KM amplifies noise as $R \to 1$. On these plates OD is the better-behaved choice, and the theoretical superiority of KM only pays off with clean, unclipped capture and a real calibration series.

Separately, the concentration response of a fluorescence-quenching plate is not linear and saturates — past a certain loading the spot is as dark as it will ever get. The origin spots on plates 3, 4 and 5 are visibly in that regime. Their areas cannot be converted to amounts even in principle.

P13. There is no ground truth

Every accuracy number in Part 3 compares one algorithm against another, because nobody has annotated these plates and no public annotated TLC image dataset exists — we searched Kaggle, Zenodo, Roboflow and Hugging Face and found none. That has two consequences: any supervised model would need a dataset built from scratch, and any claim that method X is "more accurate" than method Y on our plates currently rests on agreement, not truth. Building a modest annotated set (30–50 plates, spot centres and confirmed/absent judgements from a chemist) is the single highest-value piece of groundwork for everything in Parts 3 and 5.

Part 3 — Track A: turning the image into a graph

This part covers every route from pixels to numbers. Each entry gives what the method does, what it produced on our plates, what worked, what did not, and a verdict. The verdicts are scored on **accuracy of the conversion**, as requested, rather than on cost or deployment convenience.

3.0 The pipeline skeleton every classical method shares

All the deterministic variants share the same eight stages; they differ only in stage 4. Figure 8 walks the whole thing on plate7, the only plate whose exposure is intact.

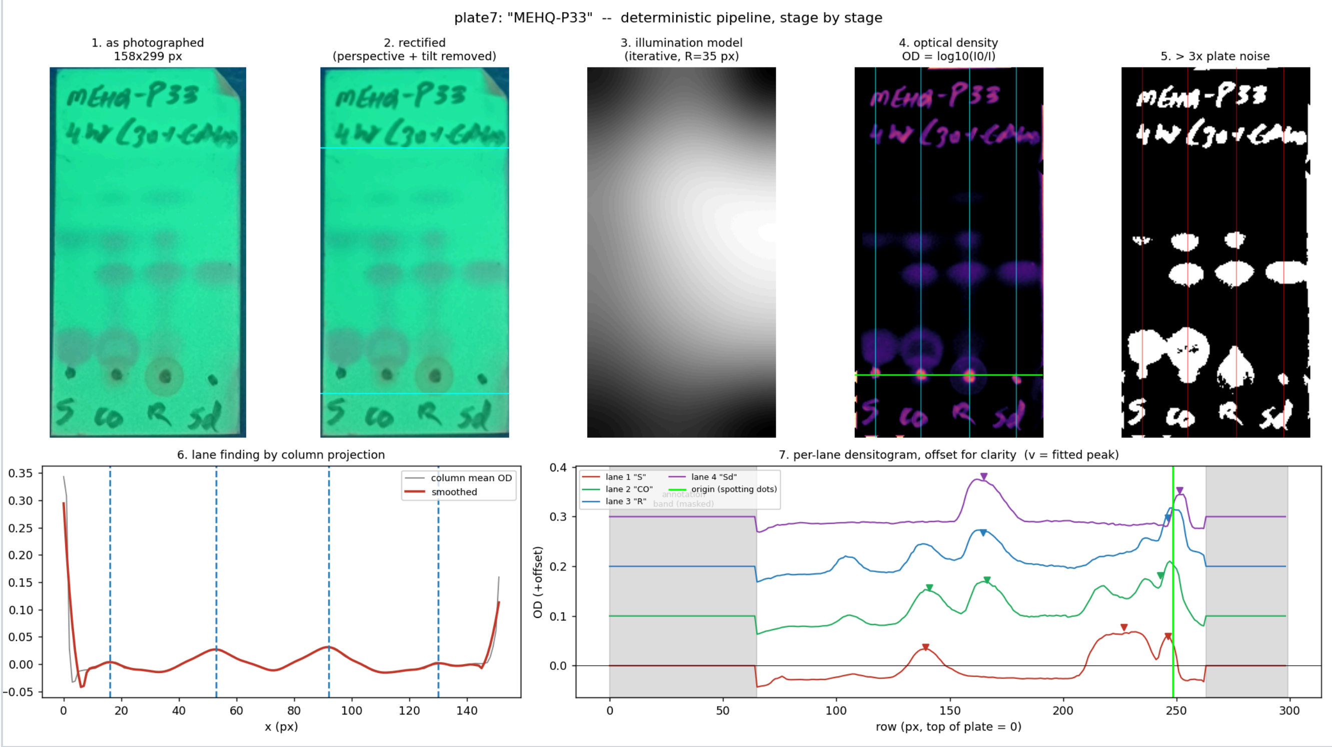

Figure 8. The deterministic pipeline stage by stage on plate7. (1) as photographed; (2) plate located and perspective-corrected, cyan lines marking the hand-supplied annotation bands; (3) the inferred illumination surface — a 15.5% brightness swing across the plate; (4) optical density, with detected lanes and the origin row from the spotting dots; (5) everything above 3× the plate's own noise — which includes the entire header, illustrating problem P11; (6) lane finding by column projection; (7) the four densitograms with fitted peaks.

3.1 The photometry engine: eight variants on identical pixels

Stage 4 — how you estimate I_0 and how you convert to a signal — is where the published methods differ. We implemented eight and ran them on the same lanes.

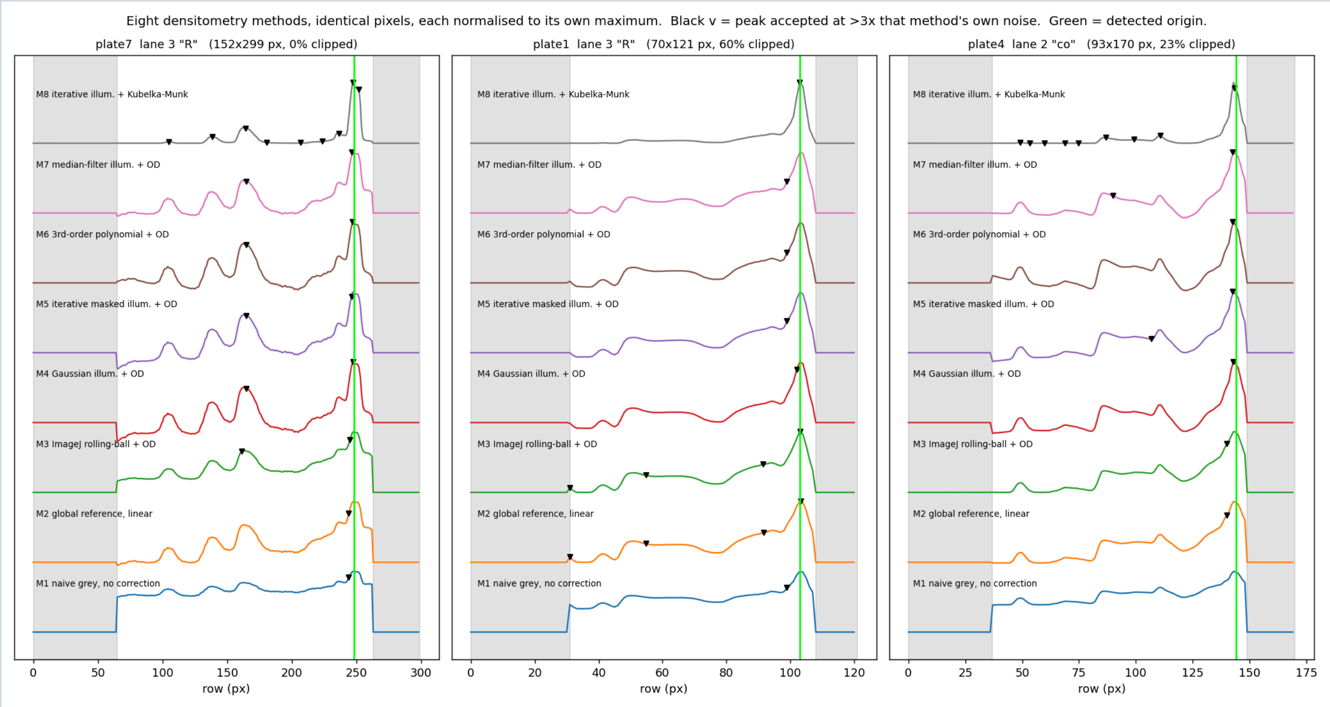

Figure 9. Eight methods, identical pixels, each normalised to its own maximum. Black triangles mark peaks accepted at 3× that method's own noise. The point of the figure is not that one curve is prettier — it is how far the accepted spot count diverges.

Method	What it does	Result on our plates	Verdict
M1 · Naive grayscale invert luminance, no correction	Convert to grey, invert, average down each lane. What you get from a two-line script.	plate7 lane 3: **1** peak. plate4 lane 2: **0**. It reports only the origin blob, because the lighting gradient swamps everything else and luminance dilutes the green signal with red/blue noise.	**Reject** Not a densitogram; a picture of the lamp.
M2 · Global reference one I_0 for the whole plate, linear difference	Take the 90th percentile of the plate as "blank" and subtract. No spatial model.	plate7: **1** peak; plate1: **4**. Cheap and surprisingly not terrible on small plates where the gradient is mild, but it charges the lighting gradient to the chemistry.	**Reject** Only defensible with a flat-field calibration frame.
M3 · ImageJ rolling-ball Sternberg background + OD	The standard in Fiji's *Analyze > Gels*. Roll a ball of radius R under the inverted surface; what it cannot reach is background.	plate7: **2** peaks, at rows 161.5 and 245.2. Same two features as M4–M7, but displaced: its first peak sits **3.3 px** from theirs, because the rolling ball erodes the leading edge of a broad band. Robust, well-understood, and the one every reviewer will recognise.	**Use** The sensible default and a good baseline.
M4 · Gaussian illumination heavy blur + OD	Blur heavily; call that the lighting. Single pass.	plate7: **2** peaks (165.0, 247.5). Fast, but spots pull the estimate down under themselves, which systematically under-reports strong spots.	**Partial** Fine for detection, biased for quantitation.
M5 · Iterative masked illumination blur, mask, replace, repeat ×3, then OD — the proposed pipeline	Blur to guess the lighting, mark pixels much darker than the guess as spots, replace them with the guess, repeat. Converges to what the plate would look like empty.	plate7: **2** peaks (164.8, 246.4) — within 0.3 px of M4, M6 and M7, and 3.3 px from M3. Its advantage over M4 is real: it does not let spots bias their own background. Its cost is the second circularity in P5 — the masking lowers the noise estimate by 1.68×.	**Use** Best of the eight, provided the noise unit is measured *before* masking.
M6 · Polynomial surface 3rd-order 2-D fit + OD	Fit a smooth polynomial to the whole plate. Physically motivated: lamp illumination really is smooth.	plate7: **2** peaks (164.9, 247.1). Fewest free parameters of any variant, hence the least opportunity to invent structure — a genuine advantage given P5.	**Use** Strongest candidate for a validated method: 10 coefficients you can print in an SOP.
M7 · Median filter large median + OD	Morphological cousin of M4, edge-preserving.	plate7: **2** peaks (165.1, 246.6). plate4: **2**, one of which (row 90) no other method found.	**Partial** Equivalent to M4/M5, no clear advantage, slower.
M8 · Kubelka–Munk $(1-R)^2/2R$ on the iterative background	The physically correct transform for a scattering layer, and what commercial HPTLC densitometers use for quantitation.	**9 peaks** on plate7 lane 3 against 2 for the other seven methods, and **9** on plate4 lane 2 against 0–2. KM diverges as reflectance approaches 1, so on faint quenching zones it amplifies noise catastrophically.	**Reject here** Right physics, wrong regime. Revisit only with clean, unclipped, calibrated capture.

The number to take from Figure 9. On one lane of one plate, eight published, defensible methods reported **1, 1, 2, 2, 2, 2, 2 and 9 spots**. On plate1 they reported 1, 4, 4, 1, 1, 1, 1 and 1. The methods do not disagree about where the strong spots are — those positions agree to about 1 px across M3–M7 — they disagree about *how many features clear the bar*. That is the practical form the phantom-spot problem takes. On the positions they do share, M4–M7 cluster to within **0.3 px** on the first peak and 1.1 px on the second; M3, the rolling-ball baseline, sits 3.3 px outside that cluster.

3.2 Approach 1 — Classical CV to structured data

What it does. The full pipeline of §3.0 with the output serialised as a spot table: lane, position, width, area, signal-to-noise, confirmed or candidate, plus plate-level flags.

What it produced. Across all seven plates: four lane centres located on 7/7 — though the lane *count* was supplied to the detector rather than discovered by it (P6); at least two origin dots on 6/7; solvent front on 0/7; and 4–12 candidate peaks per plate at 3σ *prominence* — a shape test, not an amplitude test — of which 0–7 also exceed 5σ in amplitude and are reported as confirmed. Plate2 confirmed nothing at all, which is the correct answer for a plate whose spots are that diffuse.

What works. Geometry. Plate detection, rectification, lane location and origin-dot finding are reliable, fast, and produce bit-identical output on identical input, since nothing in the chain is stochastic. It is auditable: every parameter is a number you can write into an SOP, and ICH Q2(R2) treats it as an ordinary analytical procedure.

What does not work. Everything semantic. It cannot tell writing from chemistry (P11), it cannot know how many lanes were spotted if one is empty (P6), it cannot decide whether a marginal feature is a spot, and it invents 5.5 spots on a blank plate at its own default settings (P5). Areas are meaningless on the six clipped plates.

> **Verdict.** This is the measurement layer and nothing else should compute the numbers. But it needs a semantic layer above it to tell it where to look, and it needs to report agreement rather than a single answer.

3.3 Approach 2 — CV to chromatogram (the densitogram itself)

What it does. The same field, collapsed per lane into the familiar 1-D trace. Nothing more than a presentation of Approach 1's 2-D output, but it is the artefact chemists actually want to look at.

What works. It makes the plate legible — see the bottom row of Figure 1. Tailing, streaking and shoulders that are hard to judge by eye are obvious in the trace, and two plates can be overlaid.

What does not work. The collapse is lossy and involves undocumented choices. Lane width (we used 55% of nominal lane pitch), mean versus sum, and what to do where the lane mask is partly outside the plate all change the curve. A curve with a smooth baseline looks authoritative in a way the underlying data does not justify.

> **Verdict.** Essential as an *output*, dangerous as an *intermediate*. Keep the 2-D OD field as the archived primary record; treat the densitogram as a rendering of it, and always plot the noise band alongside.

3.4 Approach 6 — Segmentation foundation models

What it does. Use a general-purpose segmenter — SAM/SAM 2, Cellpose-SAM, StarDist, Omnipose, or a trainable pixel classifier like ilastik — to discover regions, then interpret them as spots.

What we could run. Model weights could not be downloaded in this environment (huggingface.co returned 403), so SAM and Cellpose were not executed. We ran five classical and one trainable-ML segmenter as stand-ins, which is enough to make the structural point.

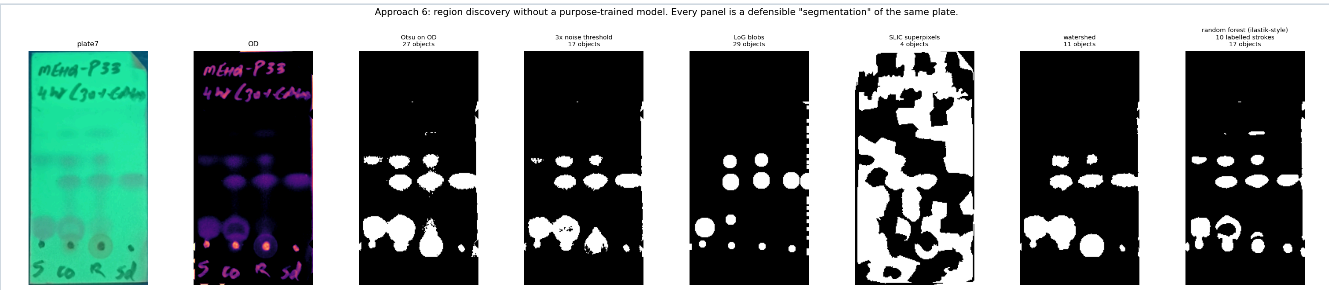

Figure 10. Six defensible segmentations of the same plate. Object counts: Otsu 27, 3σ threshold 17, LoG blobs 29, SLIC 4, watershed 11, random forest trained on six hand-drawn regions 17.

What works. The random-forest pixel classifier — the ilastik/Weka pattern — is genuinely attractive. It needed **six labelled regions** — four 5×5 patches on the origin dots and two blank bands — trains in seconds on a laptop, produces a probability map rather than a hard mask, and the model is 200 trees over 13 named filter responses, which is far easier to defend in a validation document than a 300-million-parameter transformer. Published evidence suggests fine-tuning a foundation segmenter would need 10–50 annotated plates (micro_sam, *Nature Methods* 2025, reports most of the benefit at 2–5% training fractions).

What does not work. A binary mask is the wrong output for quantitative TLC. A spot has no true boundary — its edge is a diffusion profile — so any mask truncates the tails at a learned, uncalibrated isocontour, and the integrated density depends on exactly where that cut falls. The six panels of Figure 10 differ by 7× in object count and 8× in area fraction, and all six are "correct" segmentations. Published zero-shot Dice for foundation segmenters on low-contrast, soft-edged objects runs 0.4–0.7, which is fine for "where roughly are the spots" and useless for photometry.

> **Verdict.** Use for *localisation only*, never for measurement. Of the family, the trainable pixel classifier is the best fit for us: cheapest to label, cheapest to validate. Foundation segmenters are worth a one-day experiment once we can download weights, but nobody has published SAM or Cellpose applied to gel or TLC images — we would be first, with no benchmark to compare against.

3.5 Approach 7 — Object detection trained for TLC objects

What it does. Train YOLO, RT-DETR or Faster R-CNN to emit boxes for plate, lane, spot and front.

What we ran. A COCO-pretrained YOLO11n on all seven plates, as a zero-shot baseline, with the confidence floor dropped to 0.01 — 25× below the default — specifically so that *something* would be returned. At default settings the model returns essentially nothing. The complete output:

Plate	Detections
plate1	`tv (0.379)`
plate2	`tv (0.345)`
plate3	`person (0.094), toothbrush (0.020), bird (0.011)`
plate4	`person (0.090)`
plate5	`person (0.144)`
plate6	`person (0.131)`
plate7	`person (0.060)`

What works. Nothing, without training — which is the expected and useful result. A trained detector *would* solve the two hardest geometric problems: it would find lanes regardless of whether they carry signal (P6), and it would learn to ignore handwriting (P11), because those are appearance-recognition tasks rather than signal-processing tasks.

What does not work. An axis-aligned box gives a centroid quantised to the box centre and carries no intensity information, so it cannot be the measurement. And the data cost is real: a published Mask R-CNN on 16 gel images reached only 76.6% band-localisation accuracy, while the best-in-class gel system (GelGenie, *Nature Communications* 2025) used 420 training images to reach Dice 0.82. We estimate 100–300 annotated plates for a reliable TLC detector — and no public TLC dataset exists to bootstrap from.

> **Verdict.** Worth building eventually, for lane and annotation-band localisation only, and only after 100+ plates have been annotated. Never for the numbers. Note the important precedent: GelGenie found that its neural segmenter gave *no quantification advantage* over classical background-corrected densitometry — the win was robustness and less manual work.

3.6 Approach 8 — End-to-end neural network

What it does. Image in, interpretation out, with no explicit intermediate representation.

What works. In principle it could absorb all the messiness at once. In practice we could not test it and neither has anyone else: we found **no peer-reviewed paper applying CNNs or transformers to TLC plate photographs for quantitative**

densitometry — a single unverifiable SPIE conference abstract is the entire literature.

What does not work. Three things, and they are structural rather than fixable. It needs thousands of plates with reference-method ground truth. It silently absorbs the calibration curve into its weights, so it cannot be re-calibrated without retraining. And under ICH Q2(R2), retraining a model counts as a change to the analytical procedure, which triggers revalidation — so every model update costs what the original validation cost.

> **Verdict.** Reject. Maximum data cost, maximum validation cost, zero auditability, and no evidence anywhere that it beats pixel summation over a correctly background-corrected region.

3.7 Approach 13 — Generative reconstruction to a canonical plate

What it does. Have a model redraw the messy photograph as a clean, standardised plate — super-resolution, denoising, deblurring, or a learned "restoration" step such as Cellpose 3's.

What works. The *deterministic* version of this is already in our pipeline and is valuable: perspective rectification plus normalisation to plate-height fractions is exactly "reconstruct a canonical representation", and it is invertible and auditable.

What does not work. The generative version is disqualified for measurement, for a reason that is worth stating precisely: **a learned image restoration is a non-linear, uncalibrated intensity transform**. Quantitative TLC rests on a fixed map from pixel value to optical density. Any model that "enhances" the image breaks that map in a way that cannot be undone or characterised. It will also cheerfully sharpen our JPEG ringing into crisp, convincing, non-existent spots — the same failure mode as P5, but now invisible because the output looks better than the input.

> **Verdict.** Deterministic canonicalisation: yes, already doing it. Generative restoration before photometry: reject outright. If it is ever used for the text crop, the restored image must be watermarked as non-quantitative and never fed to the densitometer.

3.8 Approach 14 — Probabilistic / multi-hypothesis output

What it does. Stop forcing one answer. Run the plate through many defensible pipelines and report how many of them agree that a spot exists at each position.

What we ran. 32 pipelines on plate7 — four background radii (12, 20, 35, 55 px) × four background models (iterative, Gaussian, rolling-ball, polynomial) × two thresholds (3σ, 4σ) — and histogrammed where they placed spots.

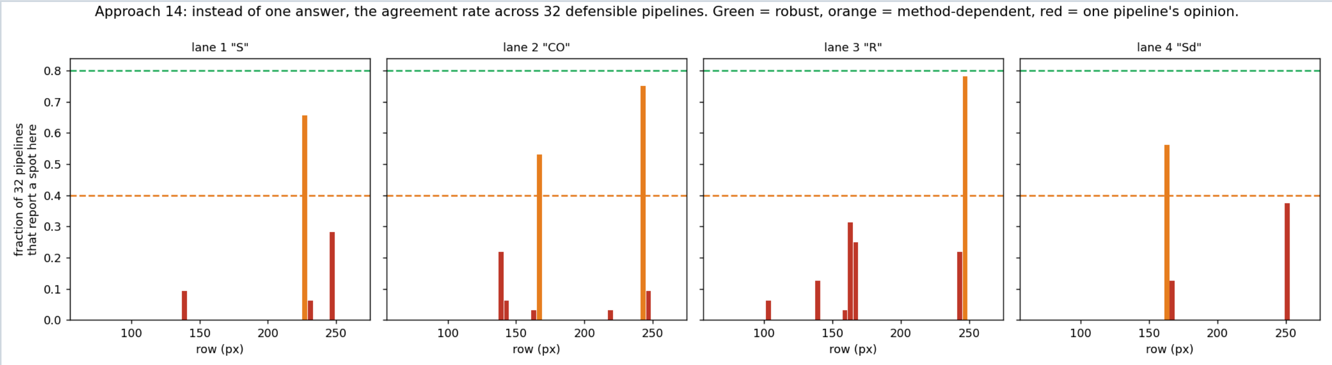

Figure 11. Instead of a spot list, an agreement profile. Bars are coloured at the 80% and 40% agreement levels. **On this plate no feature reaches 80%** — the maxima are 0.78, 0.75, 0.66 and 0.56 for lanes 3, 2, 1 and 4 — so nothing is green. That is the finding, not a rendering bug: even on our cleanest plate, no spot is found by four fifths of defensible pipelines.

What works. Everything the single-answer methods hide becomes visible — including something uncomfortable. The maximum agreement anywhere on plate7 is **0.78** (lane 3), with lanes 2, 1 and 4 peaking at 0.75, 0.66 and 0.56. Not one feature on our best plate is found by 80% of defensible pipelines. A single-answer pipeline reports those same features as facts; the ensemble reports that three quarters of methods agree and a quarter do not, which is the truth. It costs nothing but compute, needs no training data, and it is fully deterministic given a fixed config list — so it validates like an ordinary method. It also directly mitigates the problem this whole study was commissioned to investigate.

What does not work. It does not produce the single number a LIMS field wants, and someone must still choose the agreement cut-off — though that is a far more honest and stable choice than choosing a background radius. It also multiplies runtime by the number of configurations (trivial here: the full 32-pipeline sweep on one plate runs in 18 seconds as written, and about 4.5 seconds if the background is cached rather than recomputed inside the lane loop).

> **Verdict.** Strongly recommended, and the highest value-per-effort item in this entire report. It converts an unquantified risk into a reported number.

3.9 Approach 12 — Retrieval against historical plates

What it does. Reduce each plate to a signature and find its nearest neighbours in an archive: "this pattern resembles the state you had at 4 hours on P31".

What we ran. Each plate was reduced to a 4×48 vector — four lane densitograms resampled onto a common origin-anchored migration axis and normalised — and compared by correlation distance.

Plate	Nearest neighbour	Distance	Second	Distance
plate3 · MEHQ-P32 4hr	plate6 · MEHQ-P31 4hr	0.206	plate4 · P30 4hr	0.322
plate6 · MEHQ-P31 4hr	plate3 · MEHQ-P32 4hr	0.206	plate1 · P29 4hr	0.478
plate4 · MEHQ-P30 4hr	plate3 · MEHQ-P32 4hr	0.322	plate7 · P33 4hr	0.389
plate1 · MEHQ-P29 4hr	plate5 · MEHQ-P32 4+3hr	0.361	plate3	0.445
plate7 · MEHQ-P33 4hr	plate4 · MEHQ-P30 4hr	0.389	plate6	0.600
plate2 · MEHQ-P32-1 3hr	plate3 · MEHQ-P32 4hr	0.459	plate6	0.501

What works. The clustering is chemically sensible without being told anything about the chemistry: the tightest pair is P31/P32 at 4 hr, the two heavily-streaked plates (P29, P32 4+3hr) pair with each other, and P32-1 at 3 hr sits nearest its own 4 hr sibling. That is a real signal from a 20-line method with no training.

What does not work. Seven plates is not an archive. The signature inherits every problem of the underlying densitogram, so two plates can be "similar" because they were photographed with the same exposure error. And similarity is not causation — a nearest neighbour is a hypothesis for a chemist, never a conclusion.

> **Verdict.** Genuinely useful and cheap, but only once a few hundred plates with recorded outcomes exist. Build the archive now — store the OD field, not the JPEG — and this comes almost free later.

Part 4 — Track B: reading the text on the plate

This is a separate problem with a separate answer, and it should be evaluated separately — the two tracks share only the photograph. The text on our plates is a two-line handwritten header (`MEHQ-P33` / `4hr (30+GA...)`) plus a row of one- or two-character lane labels (`S co R sd`), written in marker directly on the silica, photographed under UV against a glowing green background.

We measured how large the glyphs actually are, by taking connected components of ink above 4σ within the header band. Median component height per plate: **3, 4, 5, 6, 6, 9 and 16 pixels** for plates 1–7 respectively (90th percentile 4–21 px). Only plate7 — the largest image and the only unclipped one — has strokes above 10 px.

4.1 The resolution floor, in the vendors' own words

Before choosing an engine it is worth checking whether the input is inside any of their specifications. Our measured glyph heights are **3–16 pixels, median 6**. The published minimums:

Vendor	Documented minimum text height	Their words
AWS Textract	**15 px**	"The minimum height for text to be detected is 15 pixels. At 150 DPI, this would be the same as 8 point font."
Azure AI Vision / Document Intelligence	**12 px**	"minimum height for text to be extracted is 12 pixels for a 1024 × 768 image"
Tesseract	**10 px x-height**	"Below an x-height of 10 pixels, you have very little chance of accurate results, and below about 8 pixels, most of the text will be 'noise removed'."
Google Cloud Vision	no per-character floor published	recommends 1024 × 768; "OCR requires more resolution to detect characters"

The definitive empirical evidence is TextZoom (ECCV 2020), which paired real low- and high-resolution crops of *printed* scene text: word accuracy for ASTER fell from **81.2% on 32 px text to 35.7% on 16 px text**. Our handwriting sits at or below TextZoom's low-resolution band on six of the seven plates, and handwriting is harder than print. Bicubic or Lanczos upscaling adds no information — the published ceiling for that route is 21–36% word accuracy.

4.2 What we actually measured

We ran two engines exhaustively on all seven plates. Tesseract 5.3.4 was run across **756 runs** — 108 configurations per plate (3 regions × 3 scales × 3 preprocessings × 4 page-segmentation modes) across 7 plates. EasyOCR 1.7 was run across 63 (9 per plate). The reference transcription is a frontier VLM's reading of 8× Lanczos upscales — see the caveat below.

Engine	Runs	Empty output	Mean CER, all runs	Best CER, header	Best CER, lane labels	Sample codes read correctly
Tesseract 5.3.4 (LSTM)	756	71 (9%)	0.81	0.69 avg · 0.59 best	0.44 avg · 0.11 best	**0 of 7**
EasyOCR 1.7 (CRAFT + CRNN)	63	6 (10%)	0.74	0.68 avg · 0.50 best	0.59 avg · 0.44 best	**0 of 7**
Claude Fable 5 (frontier VLM)	7	0	reference	reference	reference	**7 of 7** (self-reported)

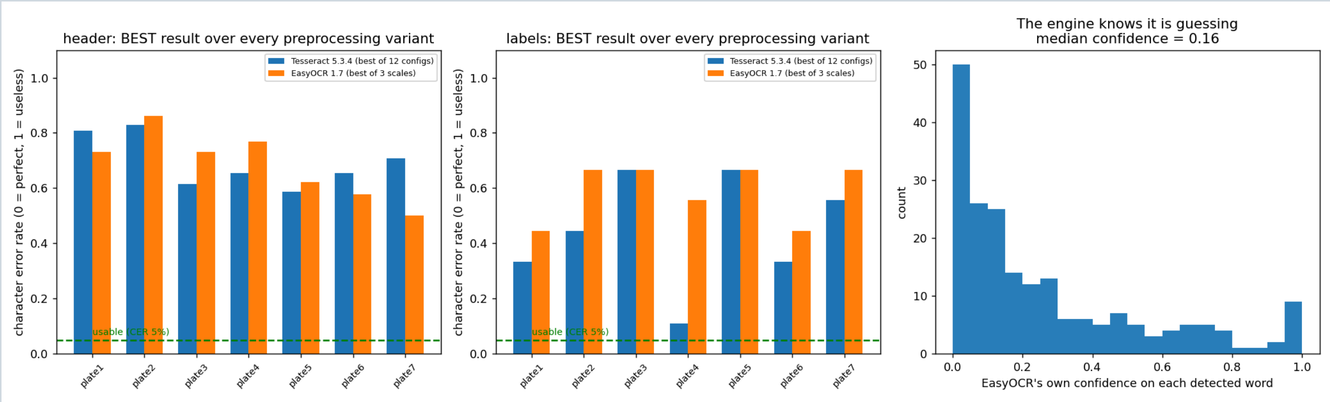

Figure 12. Best-case character error rate per plate after trying every preprocessing variant, for the header (left) and the lane labels (centre). A CER of 1.0 means the output shares nothing with the reference. Right: EasyOCR's own per-word confidence across all 203 words it emitted — median **0.16**, with **81% below 0.5** — though five words scored exactly 1.00 and were still wrong.

Verbatim outputs — the best result each engine could manage

<table>
<tr><th>Plate</th><th>Reference read</th><th>Tesseract — best of 36 header configs</th><th>EasyOCR — best of 3 header scales</th></tr>
<tr><td>plate1</td><td>MEHQ-P29 4hr (30+ GA:MeOH)</td><td>mye yoy ART LIS+ (A oy</td><td>6sti-cy 470 (306</td></tr>
<tr><td>plate2</td><td>MEHQ-P32-1 3hr (30+ EtOH:Hex)</td><td>“Cnwe- 5 ny ’ ty</td><td>"cue-Pe i sndomo</td></tr>
<tr><td>plate3</td><td>MEHQ-P32 4hr (30+ GA:MeOH)</td><td>MEna-P32- dae Ger (ane)</td><td>M{ma. 932 461 (m (t</td></tr>
<tr><td>plate4</td><td>MEHQ-P30 4hr (30+ GA:MeOH)</td><td>mena P30 4h Ger gare</td><td>Memo= Pso 4# (Mt(k</td></tr>
<tr><td>plate5</td><td>MEHQ-P32 4+3hr (30 over…)</td><td>Mewa-?52 4430 Ceve wep)</td><td>mGma-(92 4434 (8ovee 6</td></tr>
<tr><td>plate6</td><td>MEHQ-P31 4hr (30+ GA:MeOH)</td><td>Mcra-P5t She (3or.ca wer</td><td>MCma-851 4h6 (80+ 60</td></tr>
<tr><td>plate7</td><td>MEHQ-P33 4hr (30+GA…)</td><td>m&na-P 33 | tb (901 Chis</td><td>MEna-P3s N (30+ €</td></tr>
</table>

An honest caveat on the reference. The reference column is itself a frontier VLM's reading, not a transcription from the lab notebook. The CER figures therefore measure *disagreement with a frontier VLM*, not disagreement with truth, and Tejas should check the sample codes against the lab record. What the numbers do show robustly is that the Tesseract and EasyOCR outputs are not merely a different plausible reading — they are character salad that matches no plausible reading of the plate. Twenty-four of the 819 runs contained the plate's own numeric code somewhere in the output — `P32` on plate3 (20 runs), `P30` on plate4 (3), `P32-1` on plate2 (1), and EasyOCR's `p33` on plate7 — but **not one of the 819 produced a usable identifier**, because none recovered the `MEHQ` prefix and every one of those 24 buried the code in surrounding garbage. A downstream system cannot tell `MEna-P32- dae Ger (ane)` from a sample ID.

A second, more interesting caveat. The brief that commissioned this study wrote the sample prefix as `MCHQ`; reading the plates at 8× magnification, we read `MEHQ` — 4-methoxyphenol, the standard polymerisation inhibitor, which is chemically the likelier compound. Tesseract read it as `MCra`, `Mcra`, `mena` and `MEna` on different plates. At this resolution the second character is about one pixel of difference between C and E, and even a human who knows what the plate says transcribed it two ways. This is the whole problem in one character.

4.3 The engine catalogue — every named option

Prices verified 25 August 2026. "CER" figures are from the cited public benchmarks, not from our plates, except where the row says *measured here*.

Classical OCR

Engine	Licence	Size / HW	Handwriting position	Confidence output	Fit for us
Tesseract 5.5.2 (5.3.4 measured here)	Apache 2.0	~50 MB, CPU	Official FAQ: "You can, but it won't work very well, as Tesseract is designed for printed text." No official handwriting traineddata exists.	word-level 0–100	No measured CER 0.81
OCRopus / ocropy	Apache 2.0	CPU	Repository archived; superseded by Kraken.	—	Dead

Neural scene-text OCR (print-oriented)

Engine	Licence	Size / speed	Handwriting evidence	Confidence	Fit
EasyOCR 1.7.2 (1.7 measured here)	Apache 2.0	CRAFT+CRNN, CPU ok	Handwriting is still an open roadmap item — not supported.	per-box score, median 0.16 on our plates	No measured mean CER 0.74
PaddleOCR PP-OCRv5	Apache 2.0	1.75 s/img on Xeon (mobile)	Vendor's own eval: **0.581 on handwritten English vs 0.901 on printed Chinese**.	per-block	No
PaddleOCR PP-OCRv6 (Jun 2026)	Apache 2.0	1.5–34.5 M params	83.7% Hmean on its handwritten category — best in this class, but that is *detection* on clean scans.	per-block	Test it
docTR (Mindee)	Apache 2.0	4.5–23 M	No cursive model shipped.	per-element	No
Surya 2	**weights: OpenRAIL-M, free only under $5M revenue**	650 M VLM	olmOCR-bench 83.3 overall but **41.8 on old scans**.	none	Licence

Dedicated handwritten text recognition (HTR)

System	Licence	Size	Published accuracy	Trainable on our hand?	Fit
TrOCR-large-handwritten	MIT (verify)	558 M	**IAM CER 2.89%** — but IAM is 300 DPI flatbed scans of English prose with pre-segmented lines. Its RoBERTa decoder will actively regularise `MEHQ-P32` toward English words.	yes, with data	Test, expect poor
TrOCR-base-handwritten	MIT (verify)	334 M	IAM CER 3.42%	yes	Test
Kraken 7.0.2	Apache 2.0	CPU/GPU	Task-dependent; fully trainable segmentation + recognition, ALTO/PageXML output.	**yes — this is its purpose**	Best OSS route
Calamari-OCR	Apache 2.0	ensemble	<1% CER on printed Fraktur; ensemble voting cuts CER 30–50%.	yes	Alternative
PyLaia	open	—	Underpins Transkribus.	yes	Via Transkribus
Transkribus (managed)	commercial	cloud	Text Titan I: IAM CER 9.13%. Its value is **training a model on your own handwriting**.	**yes, managed**	€110–238/1k pages

Document-OCR foundation models (2024–2026) — note these are all VLMs internally

Model	Params	Licence	VRAM fp16	Benchmark / handwriting note	Fit
olmOCR-2-7B-1025 (AllenAI)	7–8 B	Apache 2.0	~16 GB	olmOCR-Bench 82.3 overall; **47.7 on old scans**. Needed dynamic temperature scaling to escape repetition loops.	Test
PaddleOCR-VL-1.6	0.9 B	Apache 2.0	~2 GB	96.33% OmniDocBench v1.5. Handles handwritten annotations "acceptably *on clean scans*".	Test
GOT-OCR 2.0	580 M	Apache 2.0	~3 GB	Doc OCR F1 0.972 EN. Trained on IAM + CASIA-HWDB2 + NorHand-v3, so it has seen handwriting — the clean-prose kind.	Test
dots.ocr	1.7 B	MIT	~3.5 GB	No published benchmark.	Unknown
DeepSeek-OCR	~3 B MoE	MIT	~8 GB	No published benchmark.	Unknown
Granite-Docling (IBM)	258 M	Apache 2.0	~0.5 GB	Explicitly "not designed for handwriting".	No
LightOnOCR-2	1 B	open	—	**0.206 on handwriting** (socOCRbench) — barely above Tesseract's 0.135.	No

Commercial cloud OCR — prices per 1,000 images, verified 25 Aug 2026

Service	Price / 1,000	Handwriting	Confidence granularity	Stated resolution limit
Google Cloud Vision `DOCUMENT_TEXT_DETECTION`	**$1.50** (→$0.60 >5M/mo)	yes — this is the handwriting path	**symbol / word / paragraph / block / page** — finest available	recommends 1024×768
Google Document AI (Enterprise OCR)	$1.50 (→$0.60)	yes	word	—
Azure Document Intelligence — Read	**$1.50** (→$0.45 committed)	yes, 9 languages, plus an **`isHandwritten` flag you can route on**	per-word 0–1	**12 px text**
AWS Textract `DetectDocumentText`	$1.50 (→$0.60 >1M)	English only	per-block 0–100	**15 px text**, ≥150 DPI advised
Mistral OCR 4.1 (public preview)	$4.00	not stated	**block-level** — the only OCR-VLM shipping explicit confidence	not stated
ABBYY FineReader Engine	no list price; ~$20–100 est. from deal data	**hand-*printed* only** (ICR, in fields and frames) — not cursive	field-level	not published
Transkribus	€110–€238	primary purpose; custom model training	platform	—
HandwritingOCR.com	$150 → ~$41.60 at volume	primary purpose	—	—

Frontier vision-language models — cost per 1,000 plate photos

Assuming a 1000×1000 px crop, a 200-token instruction and a 150-token JSON reply. Image tokenisation follows each vendor's documented rule (Anthropic: 28×28-px blocks → 1,296 tokens; OpenAI: 32×32 patches × 1.2 multiplier → 1,229; Google: flat 1,120 at default media resolution, 280 at low).

Model	Standard / 1,000	Batch API (−50%)	Note
`gemini-2.5-flash-lite` at low media resolution	**$0.11**	$0.05	Google's own docs say OCR quality saturates at *medium*; *high* "rarely improves OCR results"
`gpt-5.6-luna`	$0.23	$0.12	promotional pricing; budget ~2.5× higher after Nov 2026
`gemini-3.1-flash-lite`	$0.56	$0.28	
`gemini-3.7-flash`	$1.55	$0.78	
`claude-haiku-4-5`	$2.25	**$1.12**	good accuracy/price point
`gemini-3.1-pro-preview`	$4.44	$2.22	best measured handwriting score of any model (§4.4)
`claude-sonnet-5`	$4.49	$2.25	
`claude-opus-5`	$11.23	$5.62	

4.4 The benchmark that actually predicts our task

OCRBench and DocVQA are clean-scan printed-text benchmarks and do not predict handwriting on a UV photograph. The closest public proxy is **socOCRbench** (Feb 2026), which scores 80+ models on *degraded real-world handwriting* — handwritten census records. Normalised edit similarity, 1.0 = perfect:

Model	Handwriting score	Class
Gemini 3 Pro	**0.680**	frontier hosted
Gemini 3.1 Pro	0.649	frontier hosted
Qwen3.5-397B	0.614	large open-weight
Claude Sonnet 4.6	0.536	frontier hosted
Qwen3-VL-8B	0.480	**small open-weight — self-hostable**
Qwen3.5-2B	0.431	**small open-weight — self-hostable**
LightOnOCR-2 (1B)	0.206	OCR specialist
PP-OCRv5	0.193	classical/neural OCR
Tesseract v5	0.135	classical OCR

Two readings of this table matter for us. First, it independently confirms what we measured: a small 2–8 B VLM is **3.2–3.6× better than Tesseract** on degraded handwriting, and the gap between frontier models and classical OCR is a factor of five. Second, even the best model on earth scores 0.68 — roughly a third of characters wrong on genuinely degraded input. **Nothing here supports unattended reading of a sample identifier.**

There is one strongly encouraging counter-result. A 2026 study on recognising handwritten exam answers — single characters from a small closed vocabulary — found Gemini 3.1 Flash-Lite at **98.38% accuracy** with a 0.58% false-negative rate, beating a YOLOv5 baseline at 90.9%. Our lane labels (`S / co / R / sd`) are exactly that shape of problem. Expect near-98% on the lane labels and socOCRbench-grade results on the free-text header. That asymmetry should drive the design.

4.5 Hallucination — the risk that decides the architecture

The concern already observed at Mstack ("OCR pipelines hallucinate on our plates") is measured, severe, and well documented:

- On 250 figures with deliberately blurred labels, **GPT-5.2 fabricated a reading 96% of the time** and admitted it could not read only 8% of the time. Its admission rate *fell* when asked directly rather than open-endedly. Gemini 3.1 Pro admitted 71%; most open models 5–19%.
- On KIE-HVQA, the first benchmark for OCR hallucination on degraded documents, hallucination-free accuracy was: Gemini 2.5 Pro 35.0%, GPT-4o 30.2%, Claude 3.5/3.7 24.3–26.6%, Qwen2.5-VL-7B 21.5%. The stated root cause is

that models "default to linguistic priors rather than anchoring decisions to observable visual evidence".

- Prompting for abstention barely works: telling a model "respond UNREADABLE if unsure" moved one model's correct-refusal rate from 1.8% to only 12.4%. Fine-tuning for abstention reaches 26–34%.

Which options expose a confidence signal you can gate on — this is the decisive practical distinction:

Option	Confidence available?	Comment
Google Cloud Vision	symbol-level	Finest granularity of anything available
Azure Document Intelligence	word + isHandwritten	The handwriting flag is directly useful for routing
AWS Textract	block 0–100	
Tesseract / EasyOCR / PaddleOCR	word / box	EasyOCR's median was 0.16 with 81% below 0.5 — a useful *average* signal. But 11 words scored ≥0.90 and five scored exactly 1.00 on outputs that were still wrong (`'5 R M ('` at conf 1.00). The low mean reflects a low prior, not calibration; the high-confidence tail is confidently wrong.
Anthropic Claude API	no logprobs	Not supported
Google Gemini 3.x	no logprobs	Google confirmed logprobs are no longer returned for 3.x models
OpenAI	generally yes	Per-model support for the current lineup unverified
Self-hosted under vLLM	full token logprobs	**The strongest architectural argument for self-hosting**

The best-evidenced mitigation is **ensembling**: multi-model agreement predicts OCR correctness at 48.0–51.3 F1 versus 36.1–40.0 for VLM-as-judge, a 42% relative improvement, and five-model ensembles beat the best individual model in 91.1% of cases. At Gemini Flash-Lite prices, sampling five times costs **$0.55 per 1,000 plates** — cheaper than one Claude Haiku call. The economics make ensembling effectively free, and it is the same pattern that Approach 14 applies to the densitometry.

4.6 The option nobody lists: stop reading handwriting

Every route above is an attempt to recover information that was encoded in the worst possible medium. A 25×25 mm printed QR sticker on the plate, or a printed label from the LIMS, converts a 0.68-accuracy problem into a 1.00-accuracy problem for about two pence per plate. If the plate must stay handwritten, a **second white-light photograph of the header at full camera resolution**, taken before development, would put the text back inside every vendor's specification.

> **Verdict for Track B.** Classical and neural OCR are not close and will not get close — this is a resolution and domain problem, not a tuning problem. A frontier VLM with a constrained output schema (with `UNREADABLE` in every enum), 3–5× self-consistency sampling, and mandatory human review of any disagreement is the only route that works today, at $1–$4.50 per thousand plates. But the highest-return action by a wide margin is a capture change, and the second-highest is printing the sample ID instead of writing it.

Part 5 — The interpretation layer: what a VLM adds, and where it must not be trusted

Approaches 3, 4, 5, 9, 10, 11 and 15 in your taxonomy all sit above the measurement layer: they take the plate (and sometimes the measurements) and produce a reading, a structure, or a conclusion. They share one property that makes them qualitatively different from Part 3 — they are non-deterministic, and they will produce a fluent answer whether or not the evidence supports it. That is simultaneously their value and their hazard.

5.1 Approach 3 — VLM to a structured TLC representation

What it does. Give the model the photograph plus a schema, and ask for lanes, spots, positions, and whether a front line is visible, as JSON.

What we ran. A frontier VLM (Claude Fable 5) was shown 8× Lanczos upscales of plate6 and plate7 and asked to place the bands, before the per-plate pipeline numbers for those two plates had been computed. Its estimates were then scored against the deterministic pipeline on the same origin-anchored scale.

> **Provenance caveat, and it matters for how much weight this carries.** The VLM's reading is recorded as a literal block in `e12_vlm.py`, not as a timestamped API log, so the ordering claim above cannot be verified from the artefacts — you have our word for it. Anyone re-running this should log the model response before computing the comparison. The same applies to the reference transcriptions in `common.py`. Treat this section as a promising pilot result on two plates, not as a validated measurement.

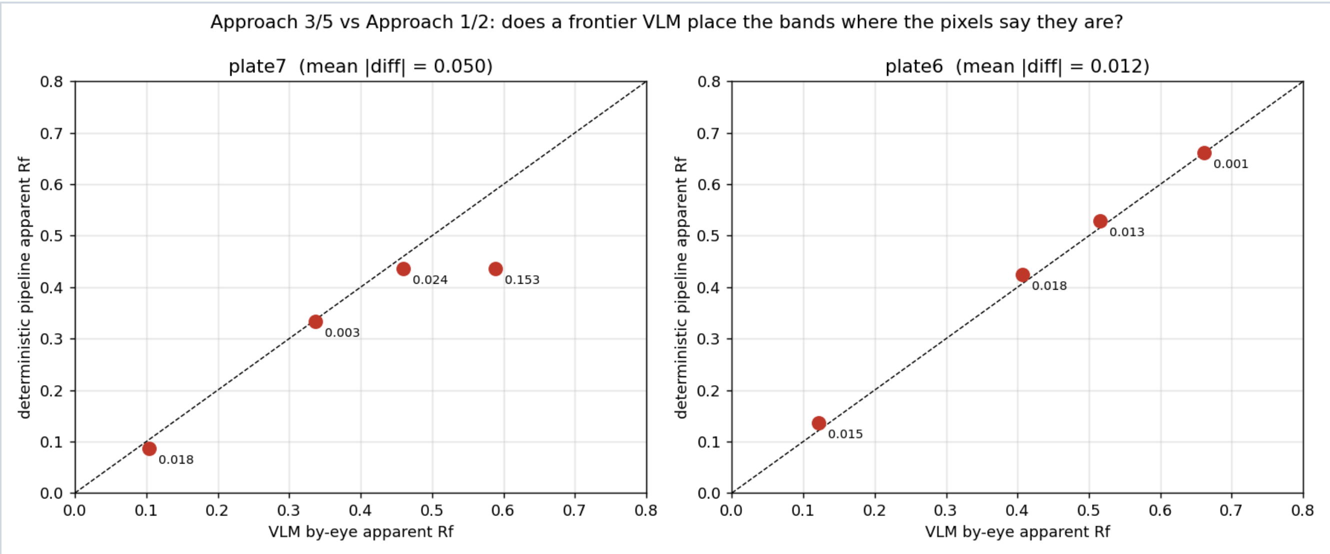

Figure 13. VLM band positions against pixel-measured band positions. Perfect agreement would lie on the dashed diagonal.

Plate	VLM apparent R_f	Pipeline apparent R_f	Difference	Comment
plate6 mean |Δ| = **0.012**	0.121	0.136	0.015	agrees
	0.407	0.425	0.018	agrees
	0.516	0.529	0.013	agrees
	0.661	0.662	0.001	agrees
plate7 mean |Δ| = **0.015** over the three matched bands; **0.050** if the unmatched band is scored against its nearest cluster	0.104	0.086	0.018	agrees
	0.337	0.334	0.003	agrees
	0.460	0.436	0.024	agrees
	0.589	—	—	**no pixel support: strongest signal at that row is 1.8σ**
plate7 — feature the VLM *missed*		0.006	—	origin residue present in all four lanes; the VLM treated it as "the origin" and did not report it as a band

What works — and this is a genuinely strong result. On plate6 the VLM placed all four bands to within 0.001–0.018 in apparent R_f. That is *inside* the ±0.02 reproducibility of manual R_f reading between human analysts. On the positions it agrees on, a frontier VLM's eyeball is as good as the densitometer. It also correctly reported `front_line_visible: false` on both plates, correctly counted the lanes, and correctly read the lane labels — three things the deterministic pipeline cannot do at all.

What does not work. Two distinct failure types, one of each, in eight bands:

- **A false positive.** The VLM reported a "very faint" band at R_f 0.589 on plate7. The pixels at that row peak at 1.8σ in the strongest lane — below any sensible threshold. It is either a real sub-threshold band or an invented one, and the VLM cannot tell you which. Note the uncomfortable nuance: the small-radius background models *did* report a peak near that row — the radius sweep in P5 finds it at R ≤ 16 px, and M8 finds it too — so this is exactly the region where CV and VLM disagree and both are guessing. The 1.8σ figure was computed for this report and sits below the pipeline's own recording threshold, so it does not appear in the output files.
- **A false negative.** The VLM did not report the origin residue present in all four lanes, because it mentally classified it as "the origin" rather than "a spot". A chemist would call that sensible; a spot table would call it a missing row.
- **No numbers.** It cannot give an area, an integrated density, or an uncertainty. Everything photometric is outside its competence.

Verdict. Excellent for structure — lane count, lane labels, whether a front exists, roughly where the bands are, which lanes share a band. Use it to *seed and constrain* the deterministic pipeline. Never let it be the source of a reported position or amount.

5.2 Approach 4 — VLM to a direct chemical interpretation

What it does. Show the plate, give the reaction context, and ask what is happening.

Worked output on plate7 (MEHQ-P33, 4 hr), verbatim in substance:

```
Four lanes: S (starting material), CO (co-spot), R (reaction), Sd (standard). All four lanes retain material at the origin, with the R lane origin spot the most intense and surrounded by a solvent halo, which suggests overloading rather than a chemical conclusion. A band at apparent Rf ≈ 0.33 appears in lanes CO, R and Sd but is weak or absent in S. A band at ≈ 0.46 appears in S, CO and R, and there is a very faint one at ≈ 0.59 in CO and R that I would not defend. Reading: the species at 0.33 is present in the reaction lane and in the standard but weak in the starting material, which is consistent with product formation; the band at 0.46 is shared between the starting-material and reaction lanes, so starting material has not obviously been consumed. The reaction looks incomplete at 4 hours. Confidence: moderate for the qualitative call, low for anything about extent of conversion, because the origin spots are overloaded.
```

What works. This is the output a chemist actually wants, expressed in the right vocabulary, and it required no pipeline at all. It also naturally hedges, which is the correct behaviour.

What does not work. Everything in that paragraph is unfalsifiable from the artefact itself. There is no audit trail from a pixel to the sentence "consistent with product formation". The published hallucination measurements (§4.5) apply in full: on degraded input, models default to plausible priors, and a reaction-monitoring plate has extremely strong priors — a model *expects* product formation, which is precisely the conclusion it is being asked to check. Nothing in the literature supports letting this stand unreviewed.

> **Verdict.** Valuable as a *draft comment* attached to a plate record, clearly marked as machine-generated, alongside the measured spot table. Never as the record itself, and never without the numbers next to it so a chemist can check the reasoning in seconds.

5.3 Approach 5 — VLM builds the lane/spot graph directly

What it does. Ask for a graph: nodes are spots, edges connect spots in different lanes that are at the same height and are therefore probably the same compound.

Worked output for plate7, built from the same recorded VLM read that §5.1 scores, so the three Part 5 outputs are mutually consistent: nodes `{S:[o, b0.10, b0.46], CO:[o, b0.10, b0.34, b0.46, b0.59?], R:[o, b0.10, b0.34, b0.46, b0.59?], Sd:[o, b0.34]}`; edges `CO.b0.34 – R.b0.34 – Sd.b0.34` and `S.b0.46 – CO.b0.46 – R.b0.46`. The `b0.59?` nodes carry the VLM's own "very faint" hedge and are the ones §5.1 shows have no pixel support.

What works. The correspondence question — "is the spot in R the same compound as the spot in Sd?" — is the actual chemical question, and it is a matching problem, which is exactly what a language model is good at. It also handles the awkward cases gracefully: a band that is present in three lanes and absent in one is a natural graph statement and an awkward table row.

What does not work. Comigration is not identity — two different compounds can share an R_f, which is why the co-spot lane exists in the first place. A graph makes the inference look more rigorous than it is. And the edges inherit every positional error from §5.1, including the invented band.

> **Verdict.** Use, but build the graph from the *measured* positions with an explicit tolerance (e.g. edges where $|\Delta R_{st}| < 0.03$), not from the VLM's reading. The VLM's contribution should be labelling the nodes, not placing them.

5.4 Approach 9 — the multimodal hybrid

What it does. Deterministic CV produces the measurements; the VLM receives both the image and the measurement table and produces the chemical conclusion.

Why this is the right shape. It is the only design in which each component does the thing it is actually good at, and — critically for validation — the numbers never pass through a non-deterministic component. A model update changes the prose, not the data. Under ICH Q2(R2) that means the measurement chain is validated once and stays valid; the learned parts affect only which pixels get looked at and how the result is described, which can be checked with a cheap geometric conformance test against a fixed reference set rather than a full accuracy/precision revalidation.

There is direct precedent. GelGenie (*Nature Communications* 2025) is exactly this architecture for gel electrophoresis — a U-Net finds the bands, deterministic pixel summation produces the numbers — and its authors found the neural half gave **no quantification advantage** over classical background-corrected densitometry. The win was robustness and less manual work. That is the honest expectation to set here too.

What does not work. The model may over-trust the table it is given. If the pipeline hands it five phantom spots from a blank region (P5), a fluent chemical story will be constructed around them. The mitigation is to pass the *agreement profile* from Approach 14, not a flat spot list, so the model sees which features are robust.

> **Verdict.** This is the recommended architecture. See Part 7.

5.5 Approach 10 — image plus external chemistry knowledge

What it does. Supply the reaction SMILES, the expected product, and reference R_f values alongside the image.

What works. It converts an open-ended question into a constrained one, which is where models are strongest (§4.4: 98.4% on closed-vocabulary handwriting versus 0.68 on open text). If the system knows the expected product's R_f in this solvent system, "is there a spot there?" becomes a yes/no with a specified location — a far easier and far more checkable question. There is also a mature literature on *predicting* R_f from structure and solvent (Xu et al., *Chem* 2022: $R^2 = 0.887$ out-of-sample on 387 compounds), which could supply that expected position automatically.

What does not work — and this is the trap. Priors are exactly what causes hallucination. Telling the model what it should see makes it more likely to report seeing it. The KIE-HVQA finding that models "default to linguistic priors rather than anchoring to observable visual evidence" is a direct warning against this approach as implemented naively.

> **Verdict.** Use the prior to **define where to measure**, then let the deterministic pipeline answer with a signal-to-noise number. Never let the prior into the model's context at the moment it is asked to read the plate.

5.6 Approach 11 — multiple plates over time

What we ran. Three of our plates form a genuine mini time-course of the same sample: MEHQ-P32-1 at 3 hr, MEHQ-P32 at 4 hr, and MEHQ-P32 at 4+3 hr. We put all three on a common origin-anchored migration axis.

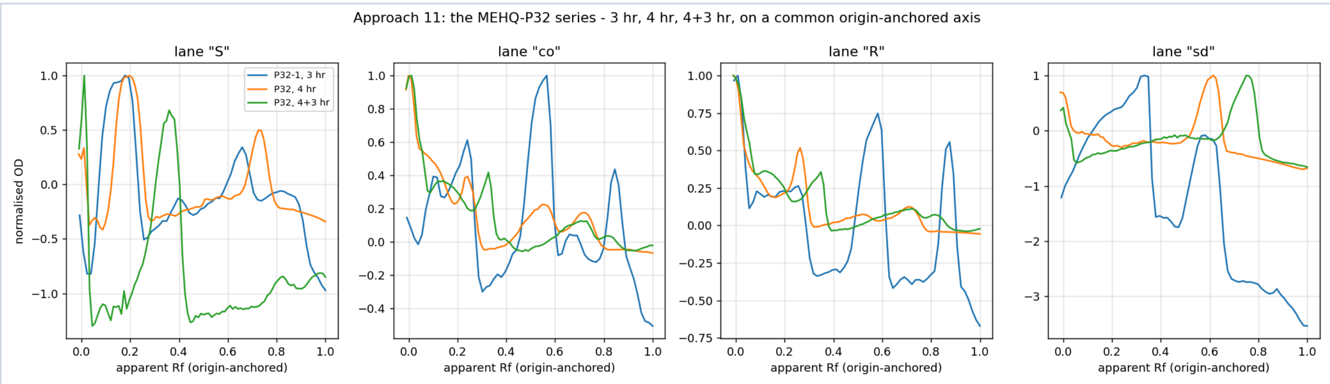

Figure 14. The MEHQ-P32 series, three time points, four lanes, common axis. The traces are not comparable — and the reason is the point of the figure.

What works, in principle. A trajectory is far more informative than any single plate, and it is robust to a lot of per-plate error: if the same band grows across three plates, that is real, whatever the absolute calibration. It also allows the strongest possible internal control, because the standard lane appears on every plate.

What did not work here. At a 4σ threshold the three plates yielded, respectively, **0, 9 and 3** confirmed spots across all lanes (at 3σ: 0, 11, 5; at 5σ: 0, 7, 2). That ordering is not chemistry. Their green-channel clipping is 14%, 35% and 24%, their sizes are 77×125, 93×166 and 112×184 px, and their tilts are 3.6°, 4.1° and 2.0°. **Trajectory analysis requires capture consistency that we do not currently have**, and no amount of normalisation recovers it — you cannot normalise away clipping.

> **Verdict.** The highest-value approach on this list *after* the capture is fixed, and worth close to nothing before. It is also the strongest argument for fixing capture, because it is what the chemists would actually get out of it.

5.7 Approach 15 — the autonomous agent

What it does. The system reads the plate, decides what it still needs, and goes and gets it — re-examining the reference lane, pulling the previous plate in the series, asking the chemist a question.

What it would look like on plate7, concretely. Read the label row → four lanes, one of them a standard. Look for a front line → none, so switch the reporting scale to R_{st} and flag it. Notice the plate touches the top image edge → flag "plate cut off, migration scale unreliable". Notice the R-lane origin spot is saturated with a solvent halo → suppress all area reporting for that lane and recommend a lower loading. Notice that the band at 0.33 appears in the standard lane → anchor R_{st} to it. Ask the chemist one question: "was a front line drawn?"

What works. Every one of those steps is a decision our fixed pipeline currently cannot make, and each one is the difference between a wrong number and an honest refusal. The agent framing is really a framing for *knowing when not to answer*, which is the thing this whole report keeps running into.

What does not work. Every step is also an opportunity for a confident wrong decision, and the failure compounds silently across steps. It is the hardest thing on the list to validate, because the execution path differs per plate.

> **Verdict.** Do not build an agent. Do steal its checklist: implement those six decisions as explicit, deterministic rules that raise flags. You get most of the value with none of the validation cost. This is the single cheapest quality improvement available.

Part 6 — The fifteen approaches, scored

Your taxonomy, with a verdict on each. "Ran it?" means we executed it on the seven MEHQ plates in this study. Scoring is on accuracy of conversion, as requested.

#	Approach	Ran it?	What it is good at	What it gets wrong	Verdict for Mstack
1	**Classical CV → structured data**	Yes §3.2	Geometry: plate, tilt, lane centres, origin dots (6/7 under the two-dot rule). Deterministic, auditable, bit-identical reruns.	Blind to meaning: cannot separate ink from analyte, cannot count empty lanes, reports 5.5 spots on a blank plate at default settings.	**Core** The measurement layer. Nothing else computes numbers.
2	**CV → chromatogram / graph**	Yes §3.3	Makes tailing, shoulders and streaking legible; allows plate-to-plate overlay; the artefact chemists want.	Lossy rendering with undocumented choices (lane width, mean vs sum). A smooth baseline looks more authoritative than the data warrants.	**Core, as output** Archive the 2-D OD field as primary.
3	**VLM → structured TLC representation**	Yes §5.1	Band positions to **0.001–0.024 apparent R_f** vs the pipeline — within human reading precision. Lane count, lane labels, front-line presence: all correct.	One invented band in eight (1.8σ support); one real feature omitted. No areas, no uncertainties, no reproducibility guarantee.	**Core, for structure** Seeds and constrains the pipeline.
4	**VLM → direct interpretation**	Yes §5.2	Produces the chemist's answer in the chemist's language, with sensible hedging, at zero engineering cost.	No audit trail. Strong priors on a reaction plate are exactly the hallucination trigger — the model expects product formation.	**Draft comment only** Always beside the numbers, always labelled.
5	**VLM → graph directly**	Yes §5.3	Lane correspondence is the real chemical question and is a natural matching task for a language model.	Comigration ≠ identity. A graph makes the inference look more rigorous than it is. Inherits positional errors.	**Yes, but build edges from measured positions**
6	**Segmentation foundation model → data**	Proxies §3.4	Random-forest pixel classifier reached a usable mask from **six labelled regions**. Foundation models need ~10–50 plates to fine-tune.	Six defensible segmenters gave 4–29 objects on one plate. A hard mask truncates the spot tails, which is where the quantitation lives. Zero-shot Dice on soft-edged objects is 0.4–0.7.	**Localisation only** Prefer ilastik-style RF; it validates.
7	**Object detection trained for TLC**	Zero-shot §3.5	Would solve lane-finding on empty lanes and the handwriting mask — both appearance tasks. Boxes are a fine *proposal* stage.	COCO YOLO11n on our plates returned "tv", "person", "toothbrush", "bird". Needs 100–300 annotated plates. A box centroid is too coarse to be the measurement.	**Later** Only after 100+ plates annotated. Never for numbers.
8	**End-to-end neural network**	No §3.6	Would absorb all messiness in one model, in principle.	No peer-reviewed precedent on TLC images exists at all. Thousands of plates needed. Absorbs the calibration into the weights. Every retrain triggers ICH revalidation.	**Reject**
9	**Multimodal hybrid**	Yes §5.4	Each component does what it is good at; numbers never pass through a non-deterministic step, so model updates do not trigger method revalidation. Direct precedent in GelGenie.	The model will build a fluent story around phantom spots if handed a flat spot list. Must be fed the agreement profile instead.	**RECOMMENDED ARCHITECTURE**
10	**Image + external chemistry knowledge**	Partial §5.5	Turns an open question into a closed one — the regime where models score 98% rather than 0.68. Expected R_f can even be predicted from structure.	Priors are the hallucination mechanism. Telling a model what to expect makes it likelier to report seeing it.	**Yes — to choose where to measure, not to inform the reading**
11	**Multiple images / temporal**	Yes §5.6	A trajectory is far more informative and far more robust than a single plate. Strongest internal control available.	Our three-point P32 series gave 0, 6 and 2 confirmed spots — driven by 14%, 35% and 24% clipping, not by chemistry. Needs capture consistency we do not have.	**High value after capture is fixed; near-zero before**
12	**Retrieval against historical plates**	Yes §3.9	Chemically sensible clustering from a 20-line method with no training: P31/P32 at 4 hr are the tightest pair (d = 0.206).	Seven plates is not an archive. Two plates can be "similar" because of a shared exposure error.	**Start archiving OD fields now; it becomes free later**

				Similarity is a hypothesis, not a finding.	
13	**Generative reconstruction**	**Deterministic only** §3.7	The deterministic form — rectify and normalise to a canonical frame — is already in the pipeline and is valuable and invertible.	Learned restoration is a non-linear uncalibrated intensity transform; it breaks the OD calibration irreversibly and will sharpen JPEG ringing into convincing spots.	**Reject the generative form** before photometry, always.
14	**Probabilistic / multi-hypothesis**	**Yes** §3.8	32 pipelines, one agreement profile. Directly converts the phantom-spot risk into a reported number. No training data, fully deterministic, seconds of compute.	Does not give the single value a LIMS field wants; someone must still pick an agreement cut-off (a far more stable choice than a background radius).	**HIGHEST VALUE PER UNIT EFFORT**
15	**Autonomous agent**	**Traced** §5.7	Its checklist is exactly the set of decisions our fixed pipeline cannot make — and each one is the difference between a wrong number and an honest refusal.	Compounding silent failure; hardest thing on the list to validate because the execution path differs per plate.	**Steal the checklist, not the architecture**

Part 7 — What each option costs

Assumption for the volume figures: **500 plates per week, 26,000 per year**. Adjust proportionally.

Option	Per 1,000 plates	Per year at 26k	Build effort	Ongoing cost driver
Deterministic CV pipeline (Parts 3.1–3.3, 3.8)	~$0 (compute)	~$0	2–4 engineer-weeks	maintenance only
+ 32-pipeline agreement ensemble	~$0	~$0	+2 days	32× compute, still seconds/plate
Tesseract / EasyOCR	$0	$0	1 day	— does not work (mean CER 0.74–0.81)
Google Vision / Azure DI / AWS Textract	$1.50	$39	2–3 days	per-image API
Gemini Flash-Lite (low media res)	**$0.11**	**$3**	3–5 days	per-image API
Gemini Flash-Lite, 5× self-consistency ensemble	$0.55	$14	+2 days	per-image API
Claude Haiku 4.5, batch API	$1.12	$29	3–5 days	per-image API
Gemini 3.1 Pro / Claude Sonnet 5, batch	$2.22–2.25	$58	3–5 days	per-image API
Mistral OCR 4.1	$4.00	$104	2 days	per-page API
Transkribus (custom HTR on our handwriting)	€110–238	€2,860–6,190	1–2 weeks + labelling	per-page credits
Self-hosted Qwen3-VL-8B on a rented A100	$0.03 (GPU time)	**$13,000** (dedicated GPU)	1–2 weeks	hourly GPU, not per image
LoRA fine-tune of a 3–8 B VLM on 500 labelled crops	one-off	~$25 GPU + ~17 chemist-hours	1–2 weeks	**the labelling, not the GPU**
Train a TLC object detector (100–300 plates)	one-off	~$50 GPU + 40–80 chemist-hours	4–8 weeks	annotation
Capture fix: fixed rig + manual exposure + front line	—	**~$200 one-off + 5 s per plate**	1 day	discipline
Printed sample-ID sticker instead of handwriting	~$31 (~3p each)	~$800	1 day	consumables — and it makes Track B a solved problem

The cost conclusion is that cost is not the issue. Every per-image option — cloud OCR, every frontier VLM, even five-fold ensembling — lands between **$3 and $104 a year** at our volume. The three expensive rows are the ones we are not recommending as the primary route (Transkribus at €2,860–6,190, a dedicated self-hosted GPU at $13,000) plus the printed sticker at ~$800, which we *are* recommending and which is cheap for what it removes. Self-hosting is *more* expensive than the API until roughly 400,000 images per month — about 185× our volume. The decision should be made entirely on accuracy, auditability and data residency, exactly as scoped. The only expensive line items are human: chemist hours for annotation, and engineer weeks for the pipeline.

Part 8 — Recommendation

8.1 Do these five things first, before choosing any model

1. **Fix the exposure.** Lock the phone's exposure (or use a fixed camera in the UV cabinet) so the green channel peaks around 220–240, never 255. Confirm with the histogram, not by eye — a clipped plate looks *better* to a human. This single change is worth more than every algorithm in this report: it moves us from one usable plate in seven to seven in seven.
2. **Draw and photograph the solvent front.** A pencil line and two seconds. Without it, R_f is undefined and the spread across defensible conventions is 0.39–0.80.
3. **Get the whole plate in frame,** with a couple of centimetres of margin. Four of seven plates currently run off the edge.
4. **Shoot at full camera resolution and keep the original file.** The images in this study are 0.01–0.05 MP; a current phone produces 12 MP or more. Something in the workflow is downsampling by roughly 250× on the largest plate and 1,300× on the smallest, and it is throwing away the entire text track. (We have not seen the originals, so this is inferred from the file sizes — worth confirming.)
5. **Report R_{st}, not R_f** — position relative to the reference spot in the standard lane on the same plate. Costs nothing, removes the largest single source of ambiguity, and is what chemists compare anyway.

8.2 The architecture to build (Approach 9, with 14 inside it)

Stage	Component	Why this component	Deterministic?
Plate location, rectification	classical CV	Works on 7/7; textbook geometry	**Yes**
Lane count, lane labels, annotation bands, "is there a front line?"	**VLM**, constrained JSON schema, `UNREADABLE` in every enum	These are semantic judgements. Every deterministic detector we tried failed: the stroke mask fired on 47–71% of pixels.	No — but it emits no numbers
Sample ID and conditions	**VLM**, 3–5× self-consistency, human review on disagreement	0/7 for Tesseract and EasyOCR. A frontier VLM produced a plausible reading for 7/7 — but that reading is also our reference, so the 7/7 is not an independent score; verify against the notebook. Ensembling improves error detection by 42% relative.	No — flagged for review
Photometry	M6 polynomial or M5 iterative, OD	Fewest free parameters (M6: 10 coefficients printable in an SOP); M5 agrees with it to 0.1 px and avoids self-bias	**Yes**
Spot detection	**32-pipeline agreement ensemble**	Turns the phantom-spot risk into a reported number instead of an unquantified hazard	**Yes**
Position and area	EMG fit to raw corrected pixels, seeded by the ensemble	Tailing is the norm on these plates; a symmetric Gaussian is the wrong function	**Yes**
Reporting scale	R_{st} against the standard lane	No front line needed; cancels run-to-run variation	**Yes**
Flags and refusals	the Approach-15 checklist as fixed rules	Clipped / cut off / no front / streaking / overloaded / <2 origin dots / lane empty	**Yes**
Chemical comment	**VLM**, given image + agreement profile + measurements	The output a chemist wants, with the numbers beside it so it can be checked in seconds	No — clearly labelled draft

The property that makes this worth the extra plumbing: **the numbers never pass through a learned component.** A model update changes which pixels get looked at and how the result is described, and can be revalidated with a cheap geometric conformance test on a fixed reference set. The measurement chain is frozen, auditable arithmetic, validated once.

8.3 What to do in what order

Week	Action	Why it comes here
1	Capture SOP: locked exposure, front line drawn, full frame, full resolution, originals kept. Re-photograph a handful of the P29–P33 plates if they still exist.	Nothing downstream is worth building on plates where 60% of the signal is clipped. Costs a day.
1–2	A chemist annotates 30–50 plates: spot centres, confirmed/absent, lane labels, header transcription from the notebook.	There is no public TLC dataset and no ground truth. This set is needed for *every* option, including "do nothing" — it is how you find out if any of this works.
2–4	Build the deterministic core: rectify → M6/M5 photometry → 32-pipeline ensemble → EMG fit → R_{st} → flags. Validate against the annotated set.	This is the part that will still be running in five years.
4–5	Add the VLM layer on Gemini Flash-Lite or Claude Haiku, constrained JSON, 3× self-consistency, human review queue for disagreements. Measure it against the annotated set.	$3–29/year. Measure before committing to a model.
5–6	Run the accuracy-vs-resolution ablation in-house: same 50 plates at five capture resolutions. No published VLM version of this curve exists.	Half a day of work that settles the capture spec with data instead of argument.
ongoing	Archive the OD field (not the JPEG) for every plate, with its flags and assumptions.	Makes Approaches 11 and 12 nearly free in a year's time, and makes any future model training possible.
later	Only if the annotated set shows the VLM layer is the bottleneck: fine-tune a small VLM (~$25 GPU + 17 chemist-hours) or train a lane/band detector (100–300 plates).	Do the cheap thing first and measure it. GelGenie's lesson is that the neural half buys robustness, not accuracy.

If the manager reads one thing. The algorithms are not the bottleneck — six of the eight densitometry methods we tested place the strong spots within a third of a pixel of one another, and a fifth within three and a half pixels. The bottleneck is that the photographs discard the information before any algorithm sees it: the green channel is clipped on six of seven plates, no solvent front was ever drawn, four plates are cut off in frame, and the files are 250× smaller than the camera can produce. A one-day change to how plates are photographed is worth more than any model on this list, and it makes almost every approach in Part 6 viable that currently is not.

Appendix A — What this study could not test

Stated plainly so the results are not over-read.

- **Model weights could not be downloaded.** `huggingface.co` and the PaddlePaddle model hosts returned HTTP 403 in this environment. TrOCR, Florence-2, PaddleOCR, GOT-OCR, olmOCR, SAM and Cellpose were therefore *not* run on our plates; everything said about them comes from published benchmarks. A ready-to-run benchmark script is included in the attached code so these can be executed on a machine with normal network access — that is a half-day of work and would close the biggest gap in this report.
- **The OCR reference transcription is a frontier VLM's reading, not the lab notebook.** CER figures measure disagreement with that reading. The sample codes should be checked against the actual records; in particular `MEHQ` vs `MCHQ` should be settled from the notebook, not from the plate.
- **Seven plates, one campaign, one operator, one camera.** Everything here describes these photographs. A second operator or a different phone could shift the capture statistics substantially.
- **No chemical ground truth.** Nobody has annotated which spots are real. All "accuracy" comparisons in Part 3 are between algorithms, not against truth.
- **The images are 250× smaller than the camera's native output.** If the originals exist, every measurement here should be repeated on them — particularly the OCR benchmark, which is currently being run below every vendor's documented minimum.
- **Quantitation was not evaluated at all,** because these plates carry no calibration series and six of seven are exposure-clipped. Nothing in this report should be read as evidence about how accurately amounts can be measured.

Appendix B — Reproducibility

Every figure and number came from code that ran on the seven image files. The attached bundle contains:

File	Contents
`tlclib.py`	Reference implementations: plate detection, rectification, seven background models, OD and Kubelka–Munk, noise estimation, stroke and text-band masks, lane finding, origin-dot and front-line detection, Gaussian and EMG peak fitting.
`common.py`	Per-plate preparation, the hand-supplied annotation bands, and the reference transcriptions.
`e1_capture.py`	Capture audit — clipping, resolution, tilt, frame overrun (Table in P1, Figures 3 and the capture bars).
`e2_walkthrough.py`	Stage-by-stage pipeline figure (Figure 8).
`e3_methods.py`	The eight densitometry methods (Figure 9).
`e4_phantom.py`	Background-radius sweep and null test A (Figure 5).
`e4b_null.py`	Null test B with real noise texture; the noise-masking bias measurement (Figure 6).
`e7_rf.py`	R_f convention sensitivity (Figure 7).
`e8_geometry.py`	Tilt and resolution studies (Figure 4).
`e10_ocr.py`	Tesseract (756 configs) and EasyOCR (63 configs) benchmark with CER scoring.
`e10b_trocr.py`	TrOCR harness — ready to run where model downloads are permitted.
`e11_alt.py`	Zero-shot YOLO, six segmentation methods, cross-plate retrieval, 32-pipeline ensemble (Figures 10, 11).
`e12_vlm.py`	VLM-versus-pipeline scoring (Figure 13).
`e13_figs.py`, `e14_problems.py`	Montage, OCR scoreboard, problem gallery (Figures 1, 2, 12).
`e15_temporal.py`	The MEHQ-P32 mini time-course and its spot counts at 3/4/5σ (Figure 14).
`out/*.json`	Raw numeric outputs behind every table.

Environment: Python 3, numpy 2.4.4, scipy 1.17.1, scikit-image 0.26.0, OpenCV 4.13.0, Tesseract 5.3.4 (leptonica 1.82.0), EasyOCR 1.7, Ultralytics 8.4.128 (YOLO11n). All runs single-threaded CPU. Model weights for TrOCR, PaddleOCR, SAM and Cellpose could not be downloaded in the environment used (see Appendix A), so `e10b_trocr.py` is supplied unrun.

Appendix C — Sources

Reference methods and validation. Hauk, Boss, Gabel, Schäfermann, Lensch, Heide, "An open-source smartphone app for the quantitative evaluation of thin-layer chromatographic analyses in medicine quality screening", *Scientific Reports* 12, 2022 (TLCyzer; ICH-validated, RSD 2.79%) · doi.org/10.1038/s41598-022-17527-y · Fichou & Morlock, "quanTLC, an online open-source solution for videodensitometric evaluation", *J. Chromatogr. A*, 2018 · ICH Q2(R2), "Validation of Analytical Procedures", adopted Nov 2023 · ISPE GAMP 5 2nd Edition, Appendix D11 (AI/ML) · FDA draft guidance, "Considerations for the Use of Artificial Intelligence to Support Regulatory Decision-Making for Drug and Biological Products", Jan 2025 · EMA, "Reflection paper on the use of AI in the medicinal product lifecycle", Sept 2024.

Image analysis. Aquilina et al., "GelGenie: an AI-powered framework for gel electrophoresis image analysis", *Nature Communications* 16, 2025 (U-Net Dice 0.82 vs watershed 0.51; *no* quantification advantage over classical) · doi.org/10.1038/s41467-025-59189-0 · Archit et al., "Segment Anything for Microscopy", *Nature Methods* 22, 2025 (fine-tuning from 1–10 images) · Pachitariu, Rariden, Stringer, "Cellpose-SAM", bioRxiv 2025 · Arganda-Carreras et al., "Trainable Weka Segmentation", *Bioinformatics* 33(15), 2017 · Sternberg rolling-ball background subtraction, as implemented in Fiji *Analyze > Gels* · Xu et al., "High-throughput discovery of chemical structure–polarity relationships", *Chem* 8(12), 2022 (R_f prediction from structure, R^2 0.887 out-of-sample) · Soliman et al., Mask R-CNN gel band localisation, *J. Medical Cyber-Physical Systems*, 2024 (76.6% on 16 gels).

OCR, HTR and VLMs. Tesseract documentation, ImproveQuality and FAQ (300 DPI guidance; 10-px x-height floor) · AWS Textract limits (15-px minimum) · Azure AI Vision / Document Intelligence Read documentation (12-px minimum) · Google Cloud Vision supported-files guidance · Wang et al., "Scene Text Image Super-Resolution in the Wild" (TextZoom), ECCV 2020 (ASTER 81.2% → 35.7%) · Li et al., TrOCR, microsoft/unilm (IAM CER 2.89% large / 3.42% base) · PP-OCRv5 and PP-OCRv6 documentation and paper (handwritten English 0.581 vs printed 0.901) · socOCRbench, Feb 2026 (Gemini 3 Pro 0.680, Qwen3-VL-8B 0.480, Tesseract v5 0.135) · "Seeing is Believing? Mitigating OCR Hallucinations in MLLMs" (KIE-HVQA), NeurIPS 2025 · Pebblous, illegible-figure-label hallucination study, 2026 (GPT-5.2 fabricates 96%) · "Consensus Entropy" multi-model agreement for OCR, arXiv 2504.11101

(+42% relative error detection) · Anthropic, OpenAI and Google Gemini pricing and vision documentation, retrieved 25 Aug 2026 · Google Cloud Vision, Azure Document Intelligence, AWS Textract, Mistral and Transkribus pricing pages, retrieved 25 Aug 2026.

Prepared 25 August 2026 for Mstack Chemicals. All experimental figures generated from the seven supplied plate photographs. Where a source could not be verified, the text says so.